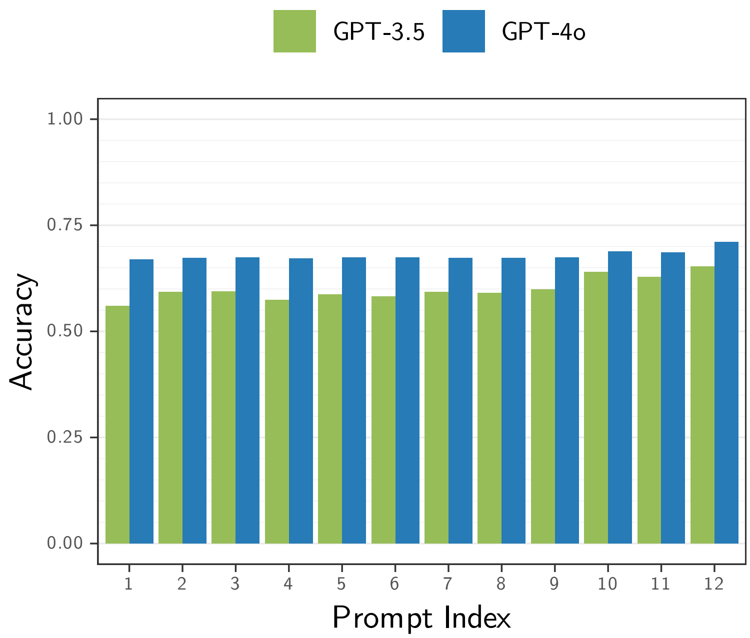
Results

Aug 12, 2024

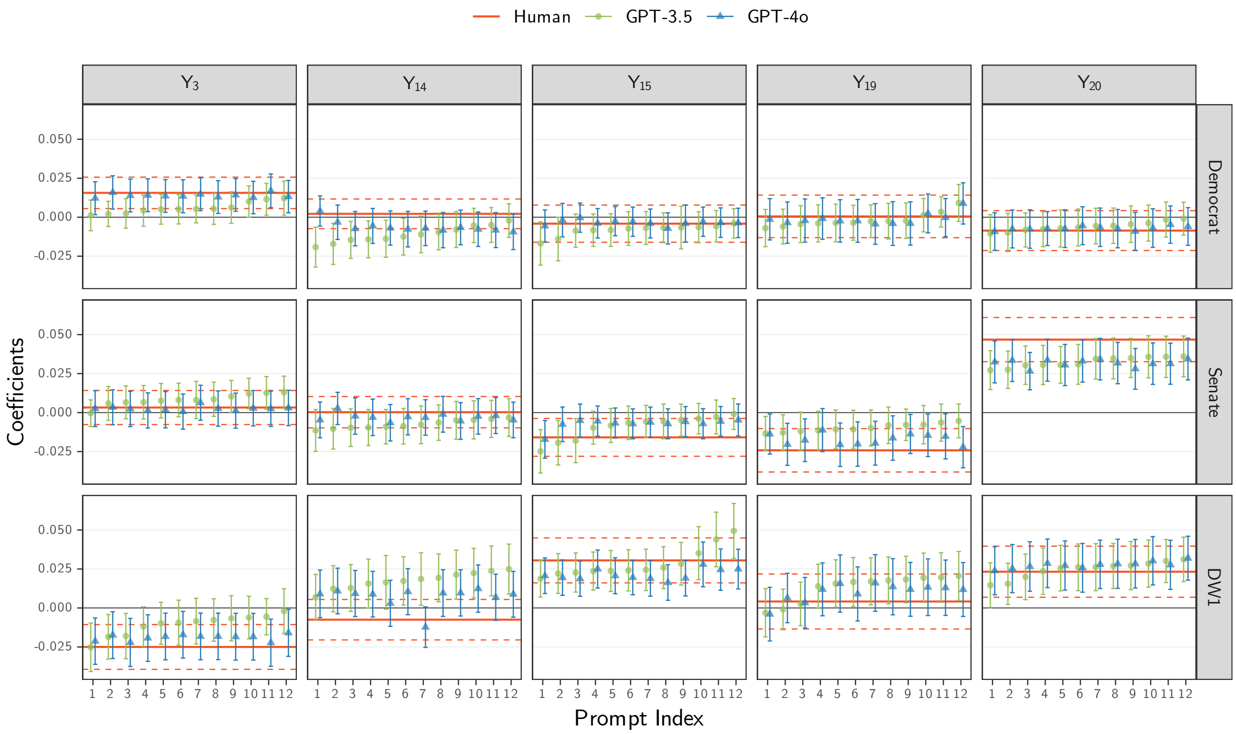
Figure 1: Accuracy of Topic Predictions vs. Prompt.



Note: Accuracy= $\frac{1}{10{,}000}\sum_{i=1}^{10{,}000} 1\{Y_{m,p}^{\text{LLM}} = Y^{\text{Human}}\}^{(i)}$

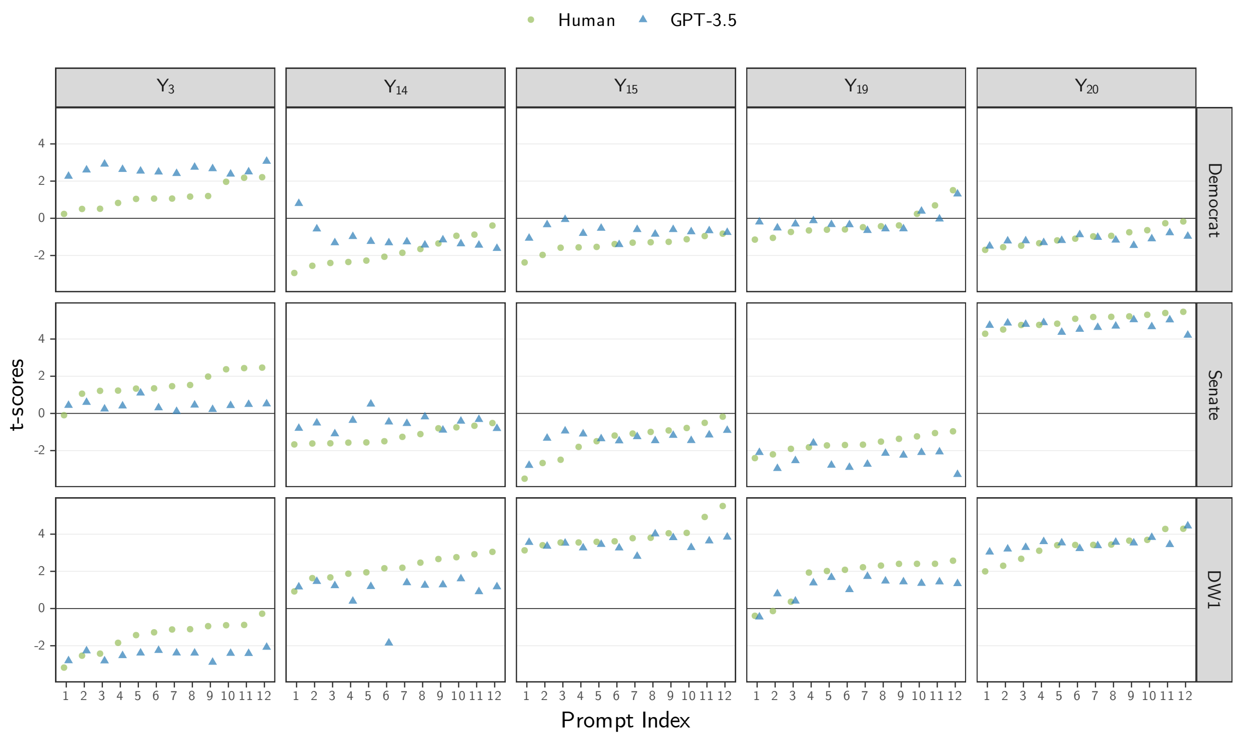
Proxy on the LHS

Figure 2: β_1 vs. Prompt. Proxy on the LHS.



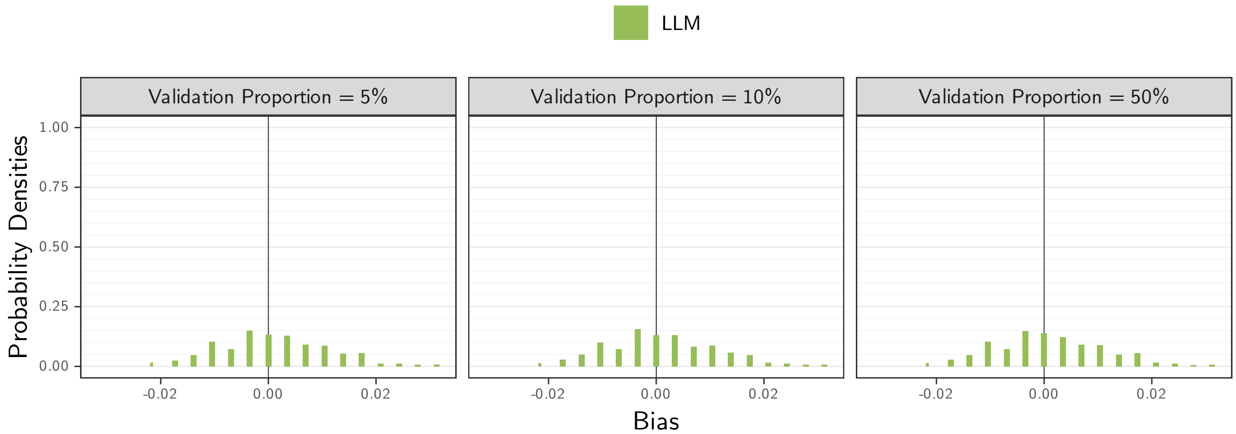
Note: The coefficients were obtained by fitting the model $Y_t = \beta_0 + \beta_1 V$ over 10,000 bills, with 95% confidence intervals computed using robust standard errors. Here, Y_t denotes the indicator for topic t, where $Y_t = 1\{Y = t\}$ for $t \in \mathcal{T}$ and $Y_{\text{Other}} = 1\{Y \notin \mathcal{T}\}$, with $\mathcal{T} = \{3, 14, 15, 19, 20\}$ is the 5 most-common human-labeled topics in the bills data. The variable V takes one of the following bills variables: Democrat, Senate, or DW1, where $extDemocrat = 1\{\text{Party} = \text{Democrat}\}$, $\text{Senate} = 1\{\text{Chamber} = \text{Senate}\}$, and DW1 is the imputed DW1 score for the bill sponsor. The regressions based on human-labeled topics (red) were estimated with $Y_t^{\text{Human}} = 1\{Y^{\text{Human}} = t\}$, while the LLM-based regressions (GPT-3.5 in green, circles; and GPT-4o in blue, triangles) were estimated with $Y_{t,m,p}^{\text{LLM}} = 1\{Y_{m,p}^{\text{LLM}} = t\}$, where $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$ denotes the LLM model and $p \in \{1, \ldots, 12\}$ denotes the prompt modifications. In each subplot, the indices for the prompts reflect the order chosen based on sorting the estimated values of the GPT-3.5 model in ascending order. The corresponding estimates for GPT-4o are aligned according to this order.

Figure 3: t-scores of β_1 vs. Prompt. Proxy on the LHS.



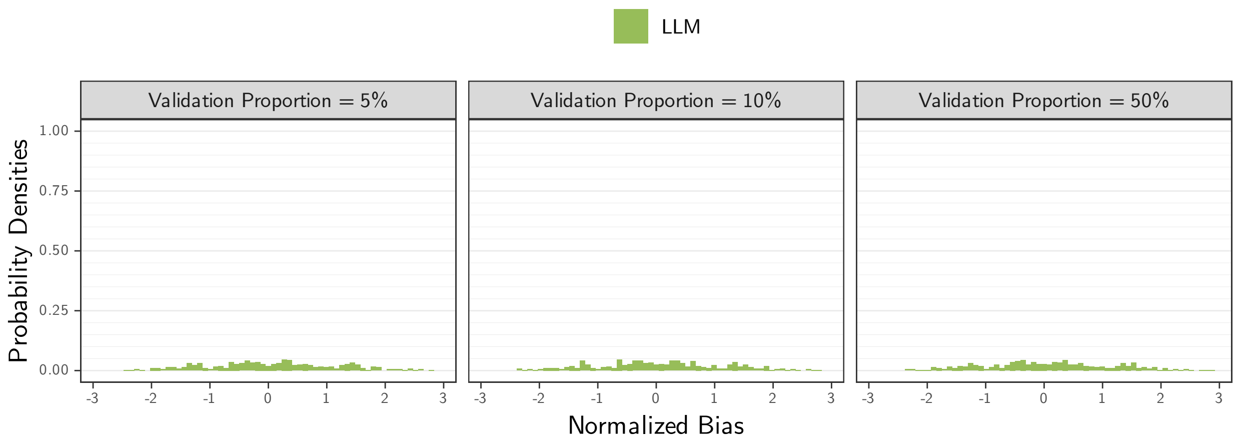
Note: The t-scores were obtained by fitting the model $Y_t = \beta_0 + \beta_1 V$ over 10,000 bills, with 95% confidence intervals computed using robust standard errors. Here, Y_t denotes the indicator for topic t, where $Y_t = 1\{Y = t\}$ for $t \in \mathcal{T}$ and $Y_{\text{Other}} = 1\{Y \notin \mathcal{T}\}$, with $\mathcal{T} = \{3, 14, 15, 19, 20\}$ is the 5 most-common human-labeled topics in the bills data. The variable V takes one of the following bills variables: Democrat, Senate, or DW1, where $extDemocrat = 1\{\text{Party} = \text{Democrat}\}$, Senate $= 1\{\text{Chamber} = \text{Senate}\}$, and DW1 is the imputed DW1 score for the bill sponsor. The regressions based on human-labeled topics (red) were estimated with $Y_t^{\text{Human}} = 1\{Y^{\text{Human}} = t\}$, while the LLM-based regressions (GPT-3.5 in green, circles; and GPT-4o in blue, triangles) were estimated with $Y_{t,m,p}^{\text{LLM}} = 1\{Y_{m,p}^{\text{LLM}} = t\}$, where $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$ denotes the LLM model and $p \in \{1, \ldots, 12\}$ denotes the prompt modifications. In each subplot, the indices for the prompts reflect the order chosen based on sorting the estimated values of the GPT-3.5 model in ascending order. The corresponding estimates for GPT-4o are aligned according to this order.

Figure 4: Bias Density of β_1 by Proportion of Validation Samples. Proxy on the LHS.



Note: The bias estimates were computed as the difference between the coefficient estimates from $N = 1000$ simulation runs, $\bar{\beta}_{t,V,\eta,m,p} = \sum_{i=1}^{N} \beta^{(i)}_{t,V,\eta,m,p}/N$, and the human-based coefficient from the entire set of 10,000 bills, $\beta^{\star}_{t,V}$, using a regression model of the form $Y_t = \beta_0 + \beta_1 V$, for topic $t \in \{3, 14, 15, 19, 20, \text{Other}\}$, variable $V \in \{\text{Democrat}, \text{Senate}, \text{DW1}\}$, validation proportion $\eta \in \{5\%, 10\%, 25\%, 50\%\}$, LLM model $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$, and prompt $p \in \{1, \ldots, 12\}$. For the LLM-based models (green), a random sample of 5,000 bills drawn from the original 10,000 was used, with LLM-based labels, $Y^{\text{LLM}}_{t,m,p} = 1\{Y^{\text{LLM}}_{m,p} = t\}$. For the other two models—human-based (red) and debiased-based (brown)—the 5,000 sample were split into two subsets: validation and primary, with the size of the validation set determined by the validation proportion, η. The results for $\eta = 25\%$ were excluded from the figures for clarity. In the human-based model, only the human labels in the validation set were used, $Y^{\text{Human}}_t = 1\{Y^{\text{Human}} = t\}$, whereas the debiased estimates use both the validation and primary sets, with Y^{Human}_t from validation set used to estimate measurement errors and $Y^{\text{LLM}}_{t,m,p}$ from the primary set used to obtain the coefficient estimates used to compute the reported bias.

Figure 5: Normalized Bias Density of β_1 by Proportion of Validation Samples. Proxy on the LHS.



Note: The normalized bias estimates were computed as the normalized difference between the coefficient estimates from $N =$ 1000 simulation runs, $\bar{\beta}_{t,V,\eta,m,p}$, and the human-based coefficient from the entire set of 10,000 bills, $\beta^{\star}_{t,V}$, divided by the sample standard deviation, $s_{t,V,\eta,m,p}$, where $s^2_{t,V,\eta,m,p} = \sum_{i=1}^{N}(\beta^{(i)}_{t,V,\eta,m,p} - \bar{\beta}_{t,V,\eta,m,p})^2/(N-1)$ and all variables are as defined previously.

Figure 6: Bias Density of β_1 by Model and Proportion of Validation Samples. Proxy on the LHS.

Figure 7: Normalized Bias Density of β_1 by Model and Proportion of Validation Samples. Proxy on the LHS.

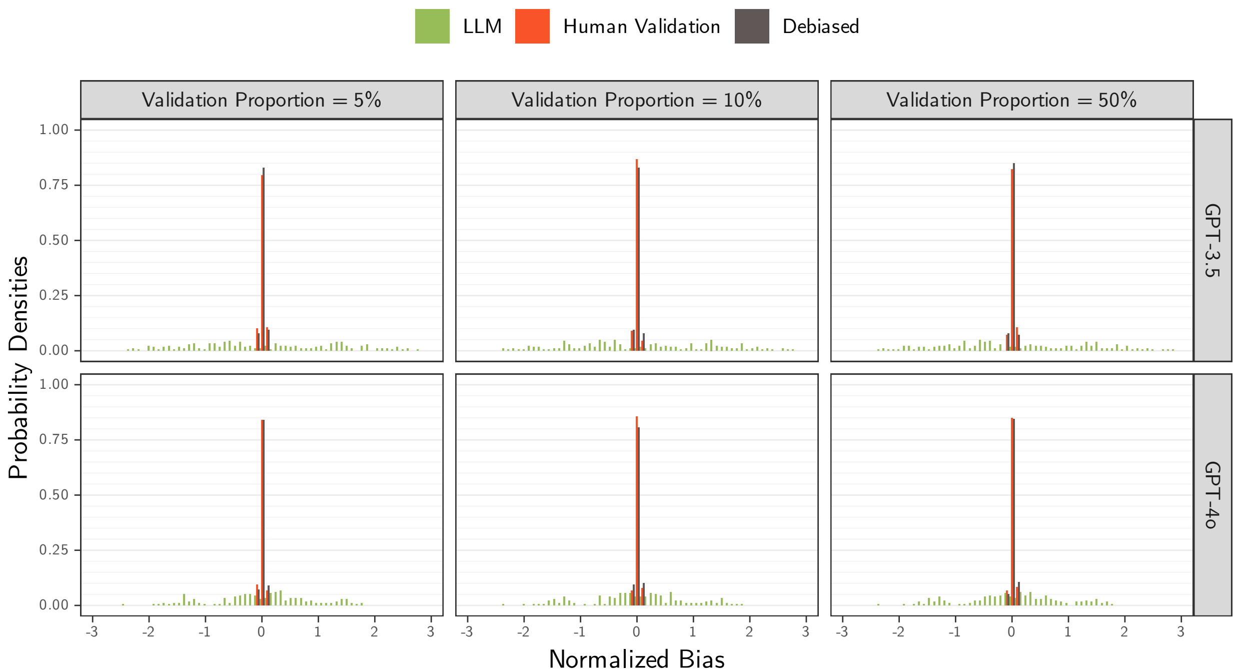


Figure 8: MSE Empirical CDF of β_1 by Proportion of Validation Samples. Proxy on the LHS.

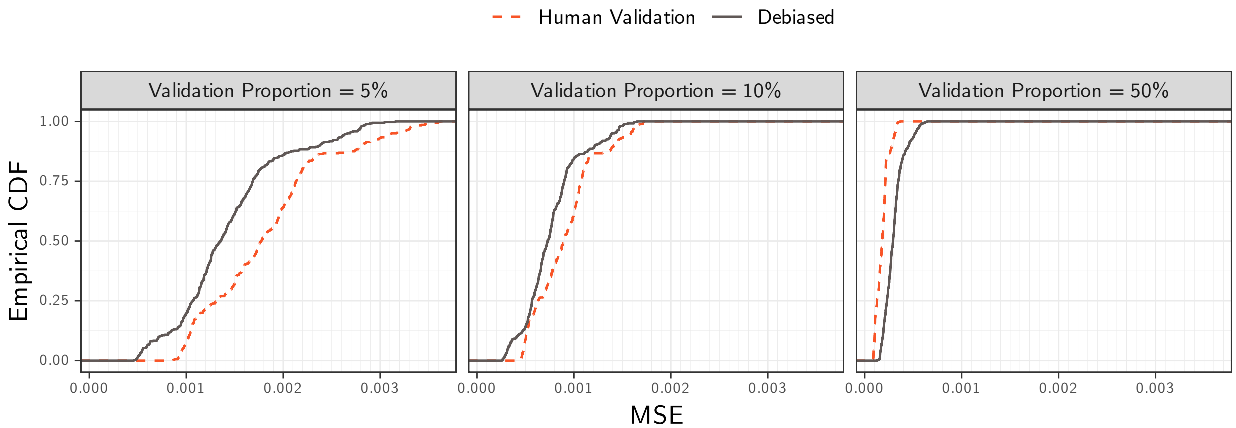


Figure 9: MSE Empirical CDF of β_1 by Model and Proportion of Validation Samples. Proxy on the LHS.

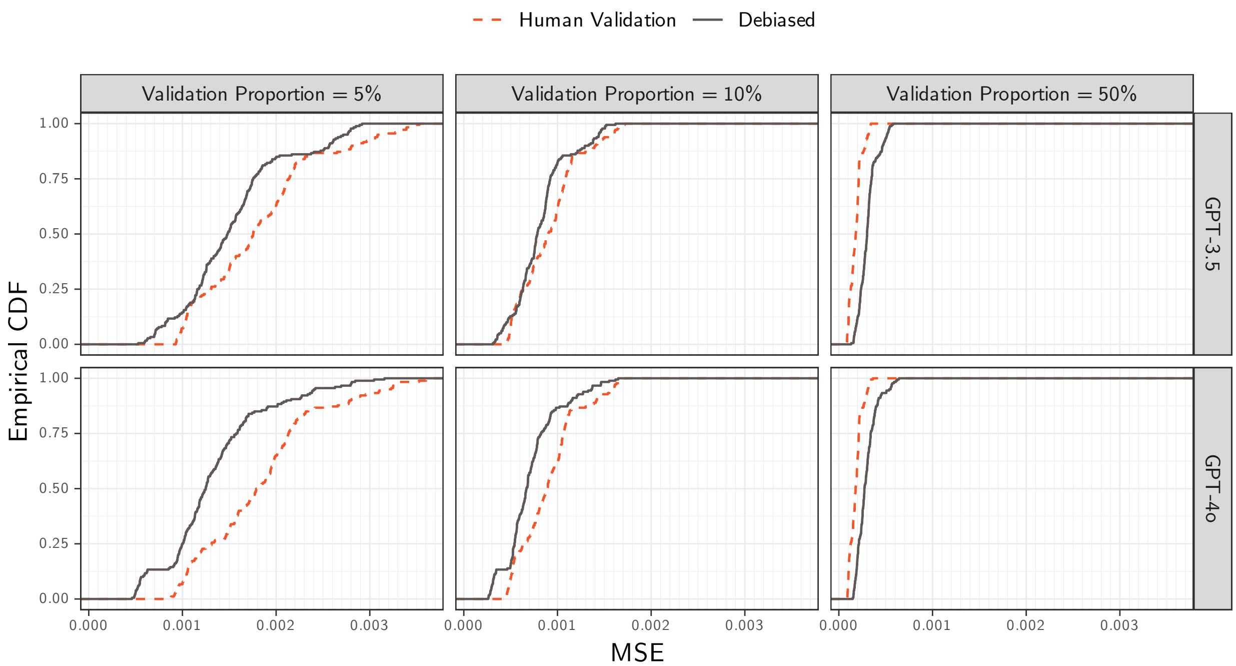


Table 1: Bias Summary Statistics of β_1 by Proportion of Validation Samples. Proxy on the LHS: $Y = \beta_0 + \beta_1 V$.

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	0.001	0.010	0.001	-0.015	0.018
5%	Human Validation	0.000	0.001	0.000	-0.002	0.002
5%	Debiased	0.000	0.001	0.000	-0.002	0.002
10%	LLM	0.001	0.010	0.000	-0.015	0.018
10%	Human Validation	0.000	0.001	0.000	-0.002	0.001
10%	Debiased	0.000	0.001	0.000	-0.002	0.001
25%	LLM	0.001	0.010	0.000	-0.014	0.018
25%	Human Validation	0.000	0.001	0.000	-0.001	0.001
25%	Debiased	0.000	0.001	0.000	-0.001	0.001
50%	LLM	0.001	0.010	0.000	-0.014	0.019
50%	Human Validation	0.000	0.000	0.000	-0.001	0.001
50%	Debiased	0.000	0.001	0.000	-0.001	0.001

Table 2: Normalized Bias Summary Statistics of β_1 by Proportion of Validation Samples. Proxy on the LHS: $Y = \beta_0 + \beta_1 V$.

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	0.071	1.099	0.067	-1.692	1.856
5%	Human Validation	0.000	0.034	0.001	-0.055	0.054
5%	Debiased	0.003	0.033	0.001	-0.055	0.061
10%	LLM	0.069	1.094	0.040	-1.722	1.869
10%	Human Validation	-0.002	0.030	-0.001	-0.055	0.048
10%	Debiased	-0.002	0.033	-0.002	-0.060	0.049
25%	LLM	0.073	1.099	0.023	-1.711	1.854
25%	Human Validation	0.002	0.031	0.001	-0.050	0.051
25%	Debiased	0.001	0.033	0.001	-0.049	0.057
50%	LLM	0.076	1.096	0.047	-1.654	1.875
50%	Human Validation	0.004	0.032	0.003	-0.051	0.056
50%	Debiased	0.000	0.031	-0.001	-0.050	0.053

Table 3: Bias Summary Statistics by Model and Proportion of Validation Samples for β_1. Proxy on the LHS: $Y = \beta_0 + \beta_1 V$.

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.002	0.012	0.000	-0.016	0.023
GPT-3.5	5%	Human Validation	0.000	0.001	0.000	-0.002	0.003
GPT-3.5	5%	Debiased	0.000	0.001	0.000	-0.002	0.002
GPT-3.5	10%	LLM	0.002	0.012	0.000	-0.016	0.023
GPT-3.5	10%	Human Validation	0.000	0.001	0.000	-0.002	0.001
GPT-3.5	10%	Debiased	0.000	0.001	0.000	-0.002	0.001
GPT-3.5	25%	LLM	0.002	0.012	0.000	-0.016	0.023
GPT-3.5	25%	Human Validation	0.000	0.001	0.000	-0.001	0.001
GPT-3.5	25%	Debiased	0.000	0.001	0.000	-0.001	0.001
GPT-3.5	50%	LLM	0.002	0.012	0.000	-0.016	0.022
GPT-3.5	50%	Human Validation	0.000	0.000	0.000	-0.001	0.001
GPT-3.5	50%	Debiased	0.000	0.001	0.000	-0.001	0.001
GPT-4o	5%	LLM	0.001	0.008	0.001	-0.013	0.015
GPT-4o	5%	Human Validation	0.000	0.001	0.000	-0.002	0.002
GPT-4o	5%	Debiased	0.000	0.001	0.000	-0.002	0.002
GPT-4o	10%	LLM	0.001	0.008	0.001	-0.014	0.014
GPT-4o	10%	Human Validation	0.000	0.001	0.000	-0.002	0.001
GPT-4o	10%	Debiased	0.000	0.001	0.000	-0.002	0.001
GPT-4o	25%	LLM	0.001	0.008	0.000	-0.013	0.014
GPT-4o	25%	Human Validation	0.000	0.001	0.000	-0.001	0.001
GPT-4o	25%	Debiased	0.000	0.001	0.000	-0.001	0.001
GPT-4o	50%	LLM	0.001	0.008	0.001	-0.013	0.015
GPT-4o	50%	Human Validation	0.000	0.000	0.000	-0.001	0.001
GPT-4o	50%	Debiased	0.000	0.001	0.000	-0.001	0.001

Table 4: Normalized Bias Summary Statistics by Model and Proportion of Validation Samples for β_1. Proxy on the LHS: $Y = \beta_0 + \beta_1 V$.

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.113	1.284	-0.023	-1.899	2.211
GPT-3.5	5%	Human Validation	0.001	0.034	0.003	-0.052	0.056
GPT-3.5	5%	Debiased	0.003	0.034	0.001	-0.055	0.066
GPT-3.5	10%	LLM	0.109	1.280	-0.005	-1.863	2.179
GPT-3.5	10%	Human Validation	-0.003	0.031	-0.001	-0.064	0.042
GPT-3.5	10%	Debiased	-0.003	0.032	-0.005	-0.053	0.049
GPT-3.5	25%	LLM	0.118	1.285	0.007	-1.837	2.264
GPT-3.5	25%	Human Validation	0.004	0.029	0.002	-0.046	0.050
GPT-3.5	25%	Debiased	0.002	0.031	0.002	-0.044	0.053
GPT-3.5	50%	LLM	0.116	1.282	-0.004	-1.864	2.172
GPT-3.5	50%	Human Validation	0.004	0.033	0.003	-0.052	0.060
GPT-3.5	50%	Debiased	0.000	0.030	0.000	-0.050	0.049
GPT-4o	5%	LLM	0.029	0.877	0.084	-1.411	1.514
GPT-4o	5%	Human Validation	-0.001	0.033	0.000	-0.055	0.051
GPT-4o	5%	Debiased	0.002	0.031	0.001	-0.055	0.054
GPT-4o	10%	LLM	0.028	0.871	0.059	-1.447	1.507
GPT-4o	10%	Human Validation	-0.001	0.030	-0.002	-0.053	0.049
GPT-4o	10%	Debiased	-0.001	0.034	0.000	-0.066	0.050
GPT-4o	25%	LLM	0.027	0.876	0.056	-1.422	1.463
GPT-4o	25%	Human Validation	0.001	0.032	-0.001	-0.050	0.052
GPT-4o	25%	Debiased	0.000	0.035	-0.001	-0.053	0.060
GPT-4o	50%	LLM	0.035	0.874	0.070	-1.441	1.510
GPT-4o	50%	Human Validation	0.004	0.031	0.003	-0.050	0.051
GPT-4o	50%	Debiased	0.000	0.031	-0.001	-0.045	0.059

Table 5: MSE Summary Statistics of β_1 by Proportion of Validation Samples. Proxy on the LHS: $Y = \beta_0 + \beta_1 V$.

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.000	0.000	0.000	0.000	0.000
5%	Human Validation	0.002	0.001	0.002	0.001	0.003
5%	Debiased	0.001	0.001	0.001	0.001	0.003
10%	LLM	0.000	0.000	0.000	0.000	0.000
10%	Human Validation	0.001	0.000	0.001	0.000	0.002
10%	Debiased	0.001	0.000	0.001	0.000	0.001
25%	LLM	0.000	0.000	0.000	0.000	0.000
25%	Human Validation	0.000	0.000	0.000	0.000	0.001
25%	Debiased	0.000	0.000	0.000	0.000	0.001
50%	LLM	0.000	0.000	0.000	0.000	0.000
50%	Human Validation	0.000	0.000	0.000	0.000	0.000
50%	Debiased	0.000	0.000	0.000	0.000	0.001

Table 6: MSE Summary Statistics by Model and Proportion of Validation Samples for β_1. Proxy on the LHS: $Y = \beta_0 + \beta_1 V$.

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.000	0.000	0.000	0.000	0.001
GPT-3.5	5%	Human Validation	0.002	0.001	0.002	0.001	0.003
GPT-3.5	5%	Debiased	0.002	0.001	0.001	0.001	0.003
GPT-3.5	10%	LLM	0.000	0.000	0.000	0.000	0.001
GPT-3.5	10%	Human Validation	0.001	0.000	0.001	0.000	0.002
GPT-3.5	10%	Debiased	0.001	0.000	0.001	0.000	0.001
GPT-3.5	25%	LLM	0.000	0.000	0.000	0.000	0.001
GPT-3.5	25%	Human Validation	0.000	0.000	0.000	0.000	0.001
GPT-3.5	25%	Debiased	0.000	0.000	0.000	0.000	0.001
GPT-3.5	50%	LLM	0.000	0.000	0.000	0.000	0.001
GPT-3.5	50%	Human Validation	0.000	0.000	0.000	0.000	0.000
GPT-3.5	50%	Debiased	0.000	0.000	0.000	0.000	0.001
GPT-4o	5%	LLM	0.000	0.000	0.000	0.000	0.000
GPT-4o	5%	Human Validation	0.002	0.001	0.002	0.001	0.003
GPT-4o	5%	Debiased	0.001	0.001	0.001	0.001	0.002
GPT-4o	10%	LLM	0.000	0.000	0.000	0.000	0.000
GPT-4o	10%	Human Validation	0.001	0.000	0.001	0.000	0.002
GPT-4o	10%	Debiased	0.001	0.000	0.001	0.000	0.001
GPT-4o	25%	LLM	0.000	0.000	0.000	0.000	0.000
GPT-4o	25%	Human Validation	0.000	0.000	0.000	0.000	0.001
GPT-4o	25%	Debiased	0.000	0.000	0.000	0.000	0.001
GPT-4o	50%	LLM	0.000	0.000	0.000	0.000	0.000
GPT-4o	50%	Human Validation	0.000	0.000	0.000	0.000	0.000
GPT-4o	50%	Debiased	0.000	0.000	0.000	0.000	0.001

Table 7: Coverage Summary Statistics of β_1 by Proportion of Validation Samples. Proxy on the LHS: $Y = \beta_0 + \beta_1 V$.

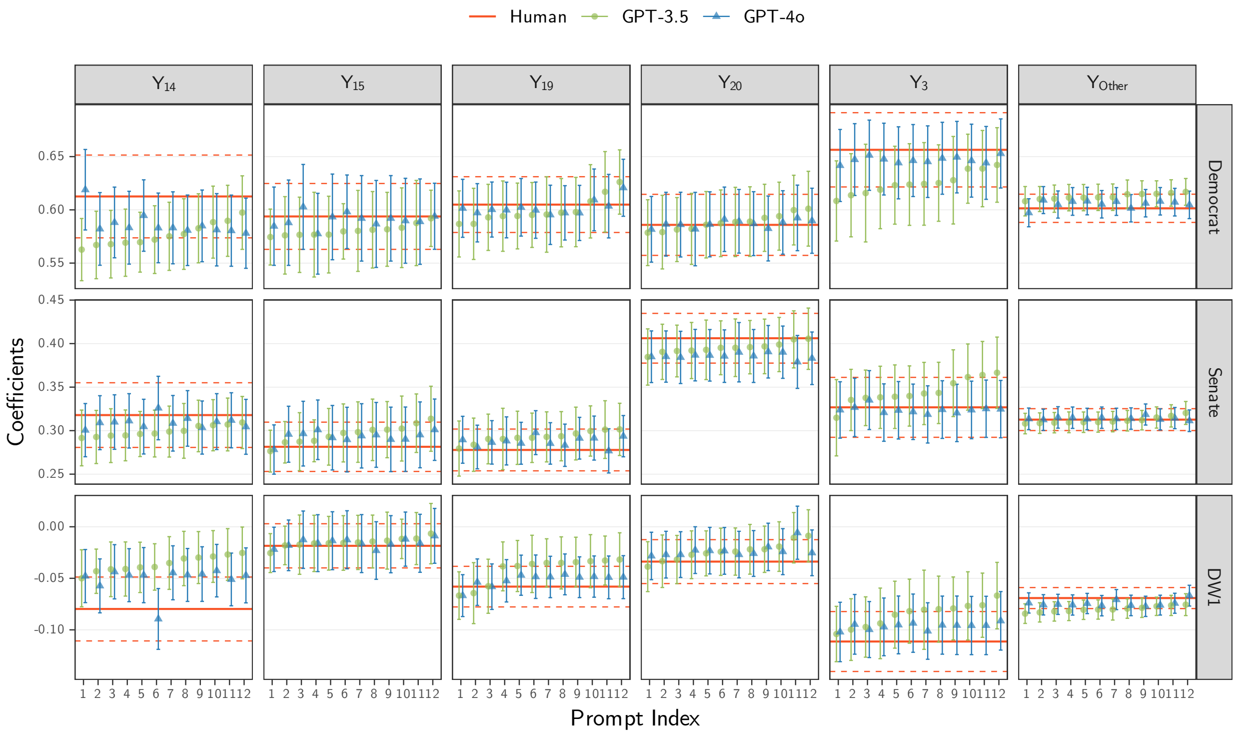
Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.811	0.163	0.889	0.461	0.951
5%	Human Validation	0.946	0.008	0.946	0.932	0.959
5%	Debiased	0.928	0.012	0.929	0.904	0.945
10%	LLM	0.810	0.165	0.889	0.460	0.951
10%	Human Validation	0.948	0.007	0.949	0.936	0.959
10%	Debiased	0.940	0.008	0.941	0.926	0.952
25%	LLM	0.809	0.164	0.887	0.465	0.952
25%	Human Validation	0.949	0.007	0.949	0.936	0.960
25%	Debiased	0.946	0.007	0.946	0.934	0.957
50%	LLM	0.810	0.164	0.887	0.463	0.950
50%	Human Validation	0.950	0.007	0.950	0.939	0.961
50%	Debiased	0.948	0.008	0.948	0.935	0.959

Table 8: Coverage Summary Statistics by Model and Proportion of Validation Samples for β_1. Proxy on the LHS: $Y = \beta_0 + \beta_1 V$.

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.761	0.187	0.819	0.381	0.945
GPT-3.5	5%	Human Validation	0.946	0.008	0.948	0.933	0.961
GPT-3.5	5%	Debiased	0.929	0.011	0.931	0.910	0.945
GPT-3.5	10%	LLM	0.759	0.189	0.812	0.383	0.949
GPT-3.5	10%	Human Validation	0.947	0.007	0.947	0.936	0.959
GPT-3.5	10%	Debiased	0.940	0.008	0.941	0.927	0.952
GPT-3.5	25%	LLM	0.760	0.188	0.815	0.362	0.945
GPT-3.5	25%	Human Validation	0.949	0.007	0.949	0.936	0.958
GPT-3.5	25%	Debiased	0.946	0.007	0.946	0.934	0.957
GPT-3.5	50%	LLM	0.760	0.190	0.815	0.369	0.950
GPT-3.5	50%	Human Validation	0.951	0.007	0.950	0.939	0.962
GPT-3.5	50%	Debiased	0.947	0.008	0.948	0.934	0.959
GPT-4o	5%	LLM	0.861	0.116	0.921	0.637	0.954
GPT-4o	5%	Human Validation	0.945	0.008	0.946	0.930	0.957
GPT-4o	5%	Debiased	0.926	0.014	0.927	0.902	0.945
GPT-4o	10%	LLM	0.860	0.117	0.919	0.642	0.952
GPT-4o	10%	Human Validation	0.949	0.007	0.949	0.936	0.959
GPT-4o	10%	Debiased	0.940	0.008	0.941	0.926	0.953
GPT-4o	25%	LLM	0.859	0.117	0.921	0.625	0.954
GPT-4o	25%	Human Validation	0.949	0.007	0.949	0.936	0.960
GPT-4o	25%	Debiased	0.946	0.007	0.946	0.934	0.958
GPT-4o	50%	LLM	0.860	0.115	0.919	0.635	0.950
GPT-4o	50%	Human Validation	0.950	0.007	0.949	0.939	0.961
GPT-4o	50%	Debiased	0.948	0.007	0.948	0.935	0.959

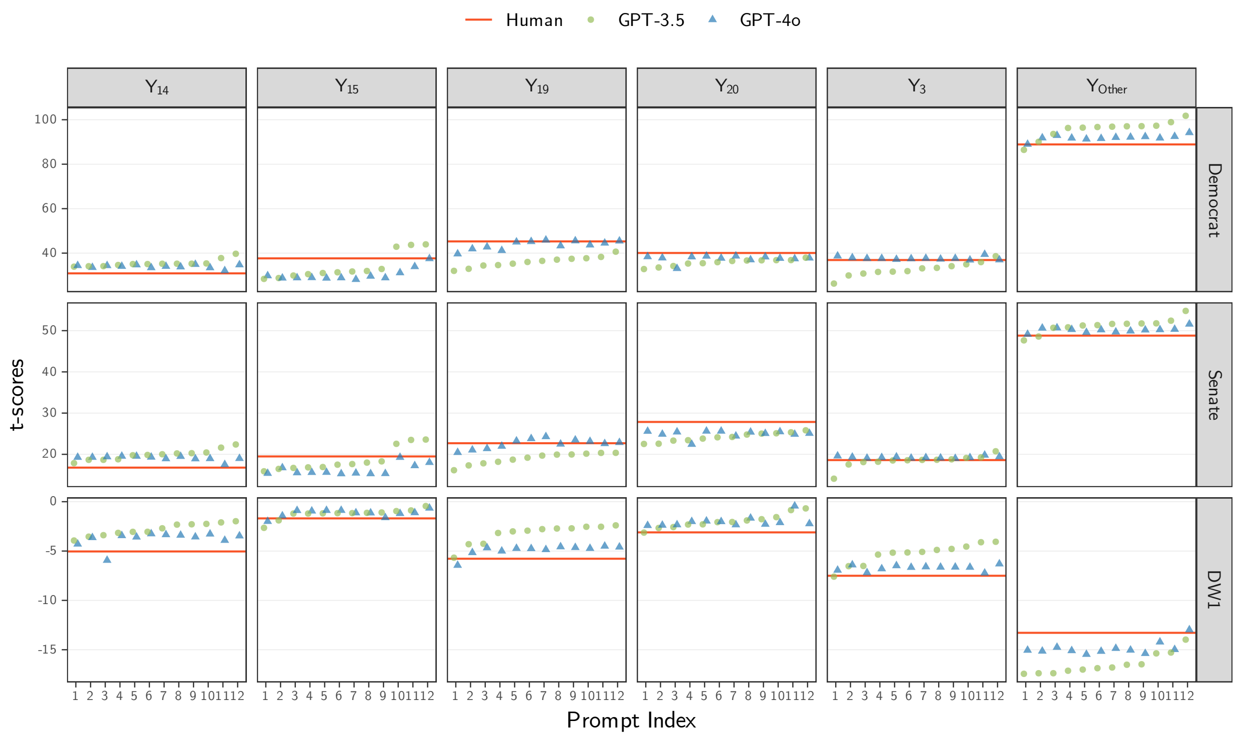
Proxy on the RHS

Figure 10: β_t vs. Prompt for all βs. Proxy on the RHS.



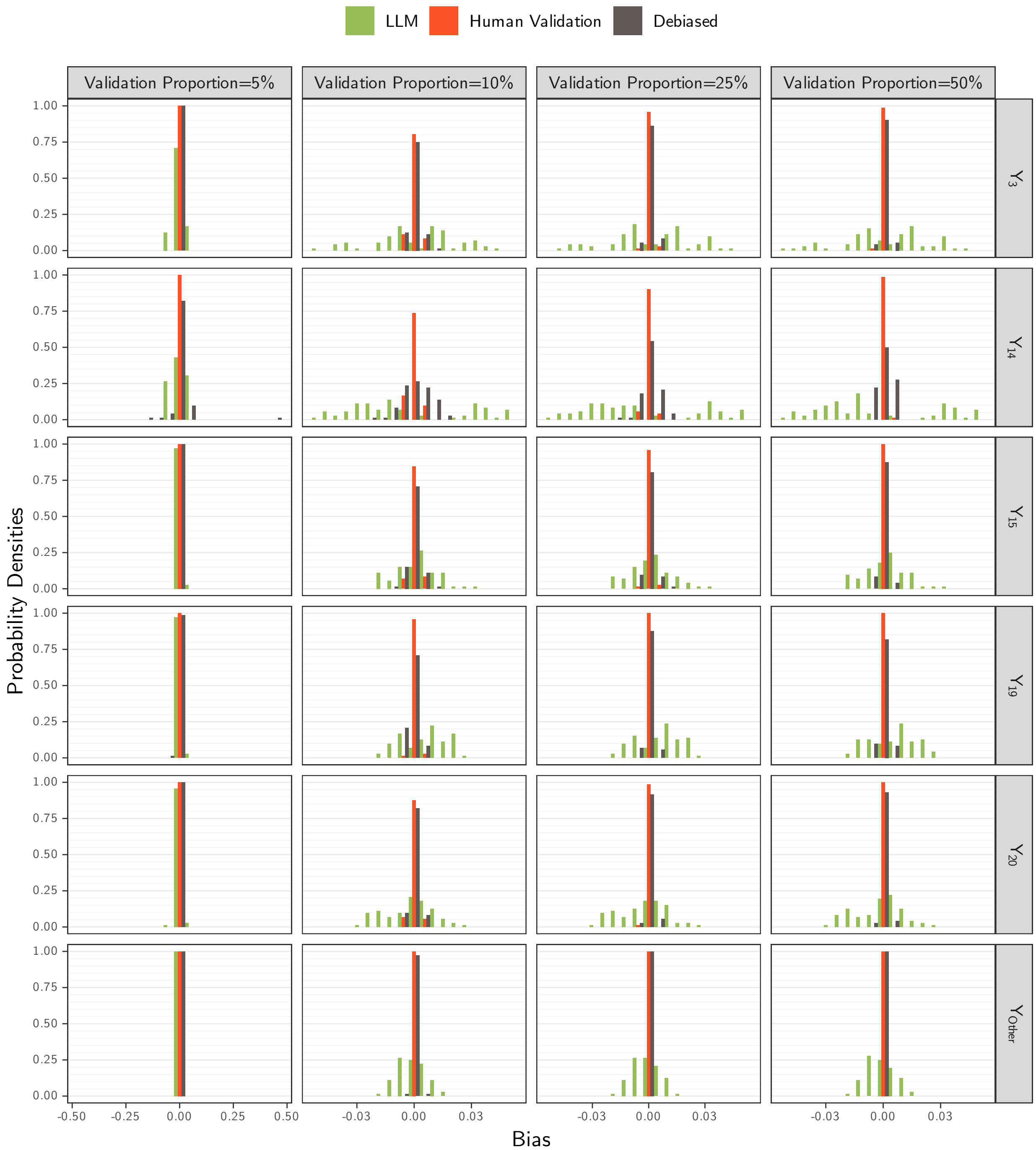
Note: The coefficients were obtained by fitting the model $V = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$ over 10,000 bills, with 95% confidence intervals computed using robust standard errors. Here, Y_t denotes the indicator for topic t, where $Y_t = 1\{Y = t\}$ for $t \in \mathcal{T}$ and $Y_{\text{Other}} = 1\{Y \notin \mathcal{T}\}$, with $\mathcal{T} = \{3, 14, 15, 19, 20\}$ is the 5 most-common human-labeled topics in the bills data. The variable V takes one of the following bills variables: Democrat, Senate, or DW1, where $extDemocrat = 1\{\text{Party} = \text{Democrat}\}$, Senate $= 1\{\text{Chamber} = \text{Senate}\}$, and DW1 is the imputed DW1 score for the bill sponsor. The regressions based on human-labeled topics (red) were estimated with $Y_t^{\text{Human}} = 1\{Y^{\text{Human}} = t\}$, while the LLM-based regressions (GPT-3.5 in green, circles; and GPT-4o in blue, triangles) were estimated with $Y_{t,m,p}^{\text{LLM}} = 1\{Y_{m,p}^{\text{LLM}} = t\}$, where $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$ denotes the LLM model and $p \in \{1, \ldots, 12\}$ denotes the prompt modifications. In each subplot, the indices for the prompts reflect the order chosen based on sorting the estimated values of the GPT-3.5 model in ascending order. The corresponding estimates for GPT-4o are aligned according to this order.

Figure 11: t-scores of each β_t vs. Prompt. Proxy on the RHS.



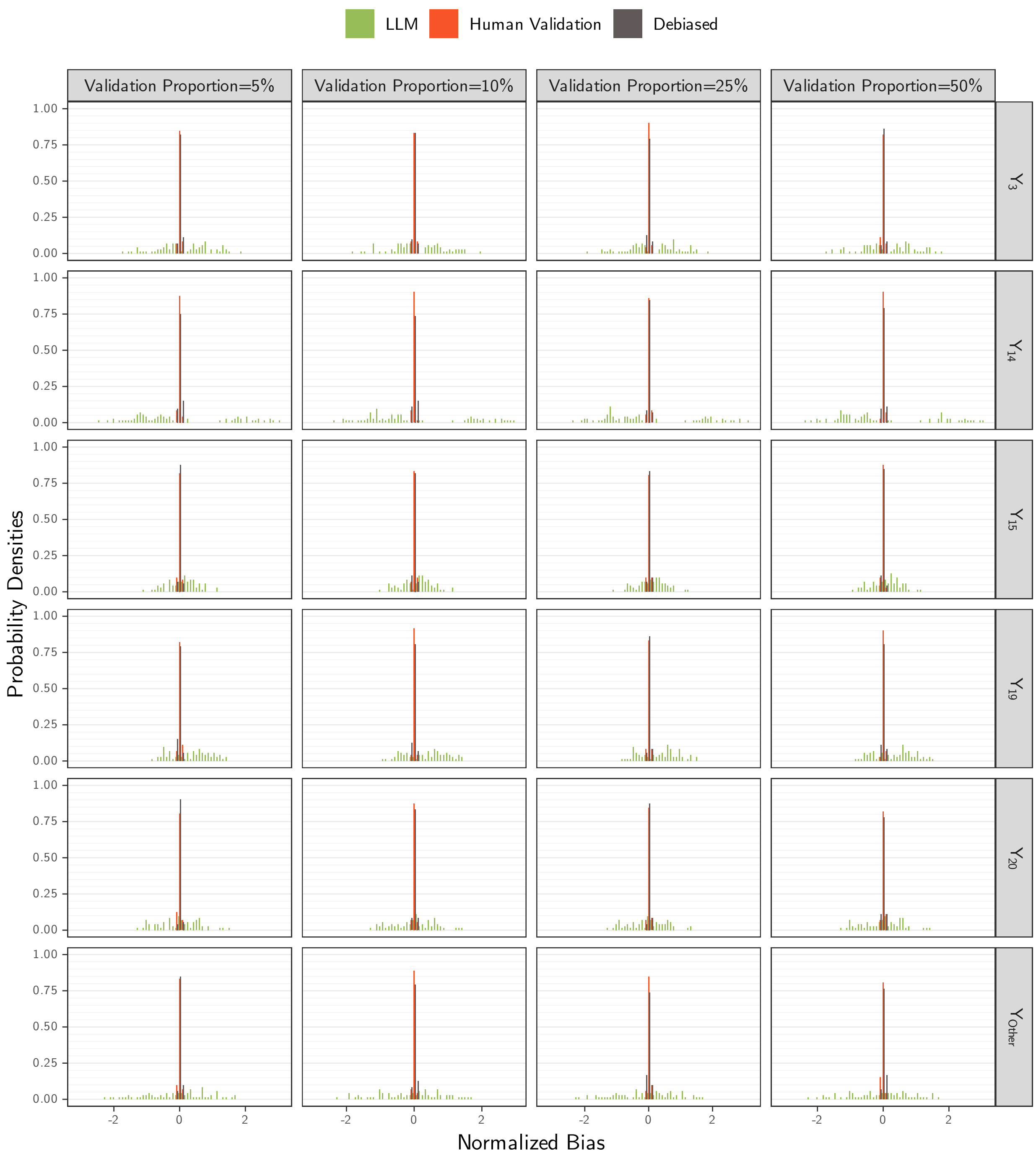
Note: The t-scores were obtained by fitting the model $V = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$ over 10,000 bills, with 95% confidence intervals computed using robust standard errors. Here, Y_t denotes the indicator for topic t, where $Y_t = 1\{Y = t\}$ for $t \in \mathcal{T}$ and $Y_{\text{Other}} = 1\{Y \notin \mathcal{T}\}$, with $\mathcal{T} = \{3, 14, 15, 19, 20\}$ is the 5 most-common human-labeled topics in the bills data. The variable V takes one of the following bills variables: Democrat, Senate, or DW1, where $extDemocrat = 1\{\text{Party} = \text{Democrat}\}$, $\text{Senate} = 1\{\text{Chamber} = \text{Senate}\}$, and DW1 is the imputed DW1 score for the bill sponsor. The regressions based on human-labeled topics (red) were estimated with $Y_t^{\text{Human}} = 1\{Y^{\text{Human}} = t\}$, while the LLM-based regressions (GPT-3.5 in green, circles; and GPT-4o in blue, triangles) were estimated with $Y_{t,m,p}^{\text{LLM}} = 1\{Y_{m,p}^{\text{LLM}} = t\}$, where $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$ denotes the LLM model and $p \in \{1, \ldots, 12\}$ denotes the prompt modifications. In each subplot, the indices for the prompts reflect the order chosen based on sorting the estimated values of the GPT-3.5 model in ascending order. The corresponding estimates for GPT-4o are aligned according to this order.

Figure 12: Bias Density Plots for all βs by Proportion of Validation Samples. Proxy on the RHS.



Note: The bias estimates were computed as the difference between the coefficient estimates from $N = 1000$ simulation runs, $\bar{\beta}_{V,\eta,m,p} = \sum_{i=1}^{N} \beta^{(i)}_{t,V,\eta,m,p}/N$, and the human-based coefficient from the entire set of 10,000 bills, $\beta^{\star}_{t,V}$, using a regression model of the form $V = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$, for topic $t \in \{3, 14, 15, 19, 20, \text{Other}\}$, variable $V \in \{\text{Democrat}, \text{Senate}, \text{DW1}\}$, validation proportion $\eta \in \{5\%, 10\%, 25\%, 50\%\}$, LLM model $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$, and prompt $p \in \{1, \ldots, 12\}$. For the LLM-based models (green), a random sample of 5,000 bills drawn from the original 10,000 was used, with LLM-based labels, $Y^{\text{LLM}}_{t,m,p} = 1\{Y^{\text{LLM}}_{m,p} = t\}$. For the other two models—human-based (red) and debiased-based (brown)—the 5,000 sample were split into two subsets: validation and primary, with the size of the validation set determined by the validation proportion, η. In the human-based model, only the human labels in the validation set were used, $Y^{\text{Human}}_t = 1\{Y^{\text{Human}} = t\}$, whereas the debiased estimates use both the validation and primary sets, with Y^{Human}_t from validation set used to estimate measurement errors and $Y^{\text{LLM}}_{t,m,p}$ from the primary set used to obtain the coefficient estimates used to compute the reported bias.

Figure 13: Normalized Bias Density of β_1 by Proportion of Validation Samples. Proxy on the LHS.



Note: The normalized bias estimates were computed as the normalized difference between the coefficient estimates from $N = 1000$ simulation runs, $\bar{\beta}_{t,V,\eta,m,p}$, and the human-based coefficient from the entire set of 10,000 bills, $\beta^{\star}_{t,V}$, divided by the sample standard deviation, $s_{t,V,\eta,m,p}$, where $s^2_{t,V,\eta,m,p} = \sum_{i=1}^{N}(\beta^{(i)}_{t,V,\eta,m,p} - \bar{\beta}_{t,V,\eta,m,p})^2/(N-1)$ and all variables are as defined previously.

Figure 14: Bias Density of β_3 by Model and Proportion of Validation Samples. Proxy on the RHS.

Figure 15: Bias Density of β_{14} by Model and Proportion of Validation Samples. Proxy on the RHS.

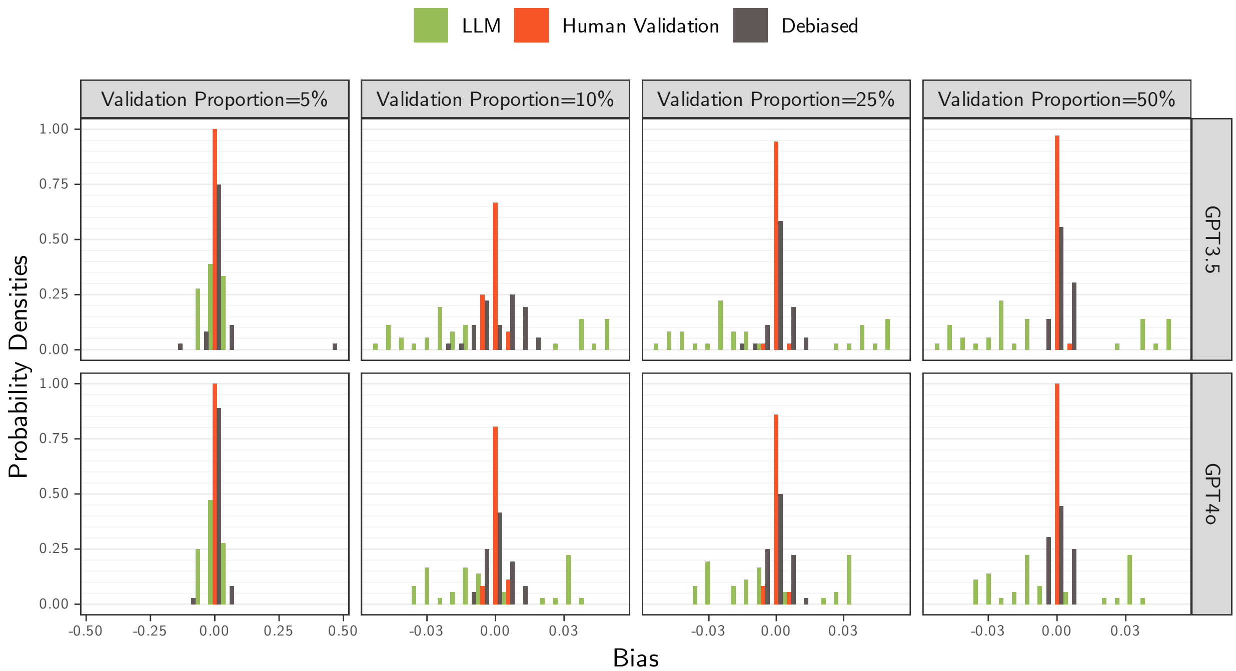


Figure 16: Bias Density of β_{15} by Model and Proportion of Validation Samples. Proxy on the RHS.

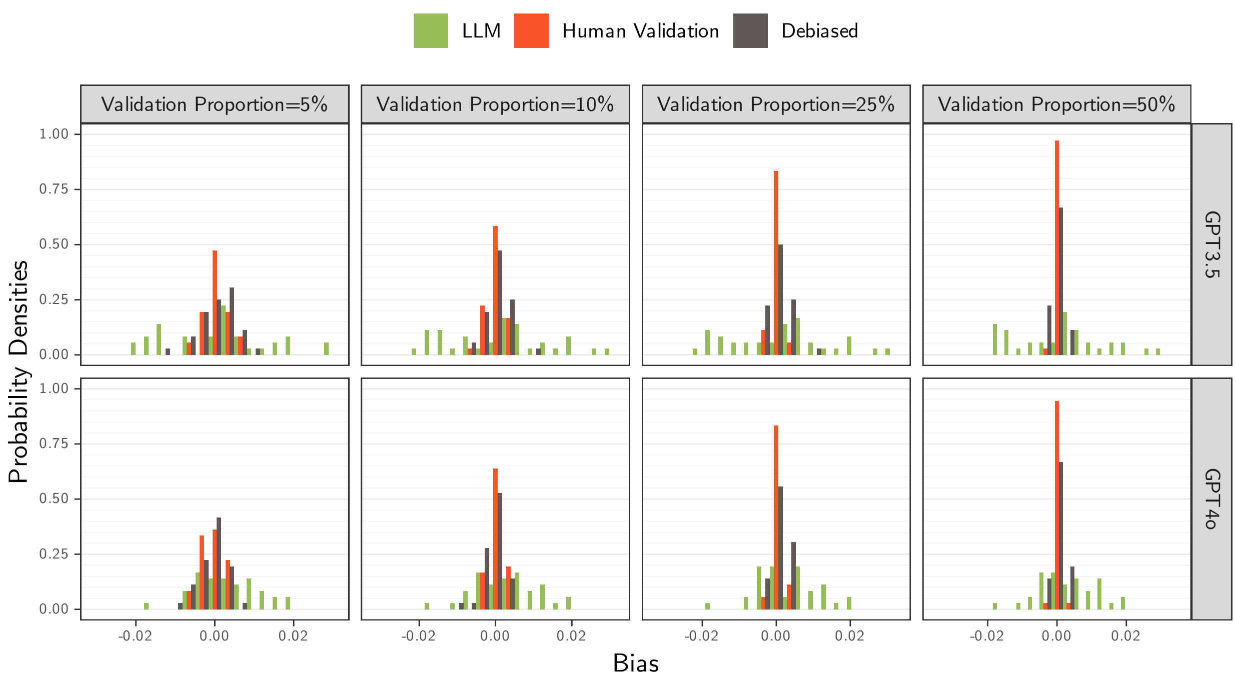


Figure 17: Bias Density of β_{19} by Model and Proportion of Validation Samples. Proxy on the RHS.

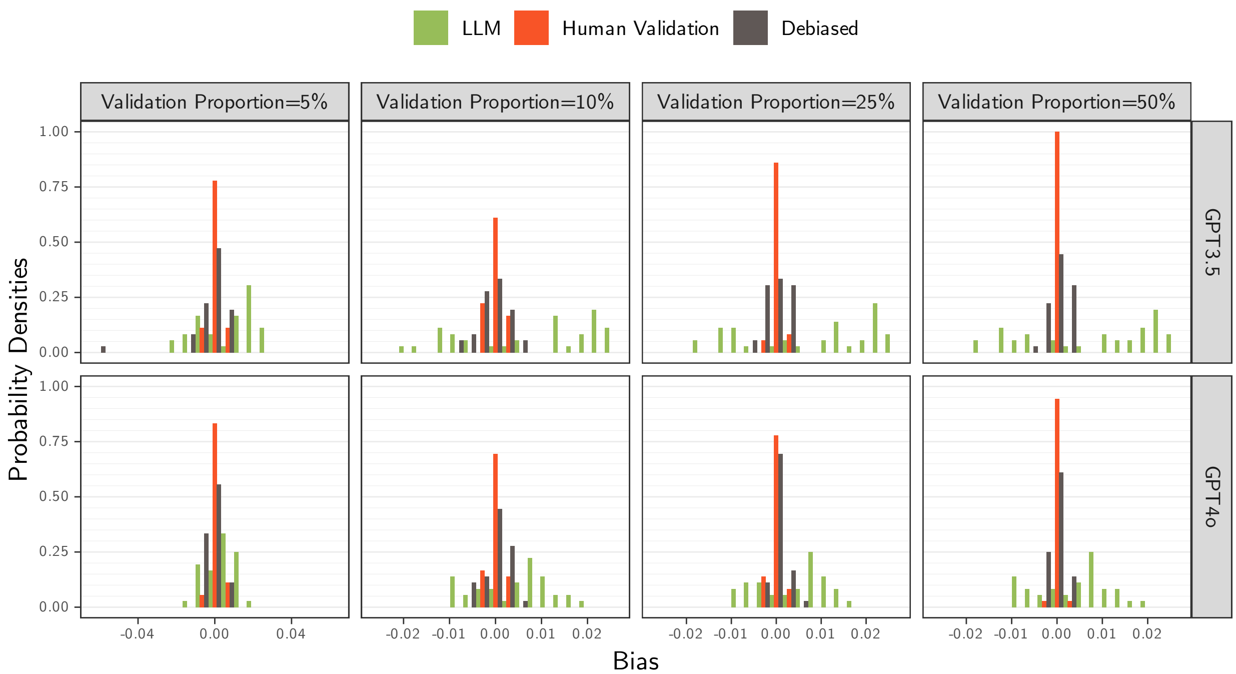


Figure 18: Bias Density of β_{20} by Model and Proportion of Validation Samples. Proxy on the RHS.

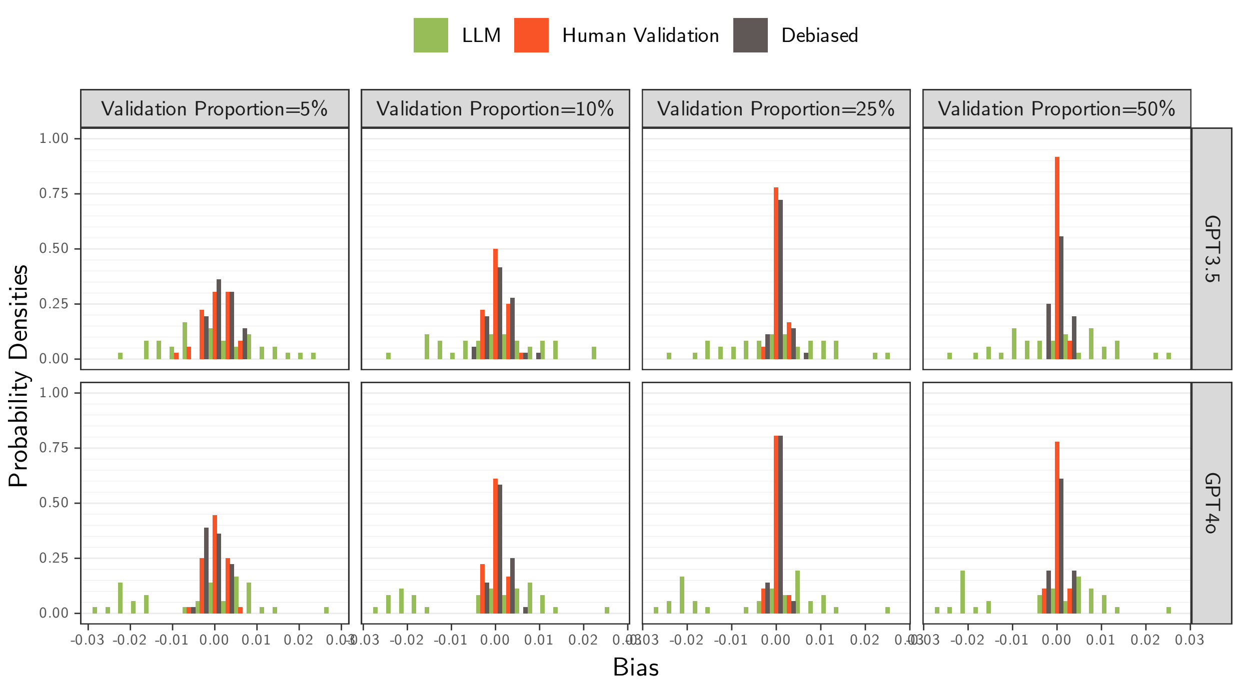


Figure 19: Bias Density of β_{Other} by Model and Proportion of Validation Samples. Proxy on the RHS.

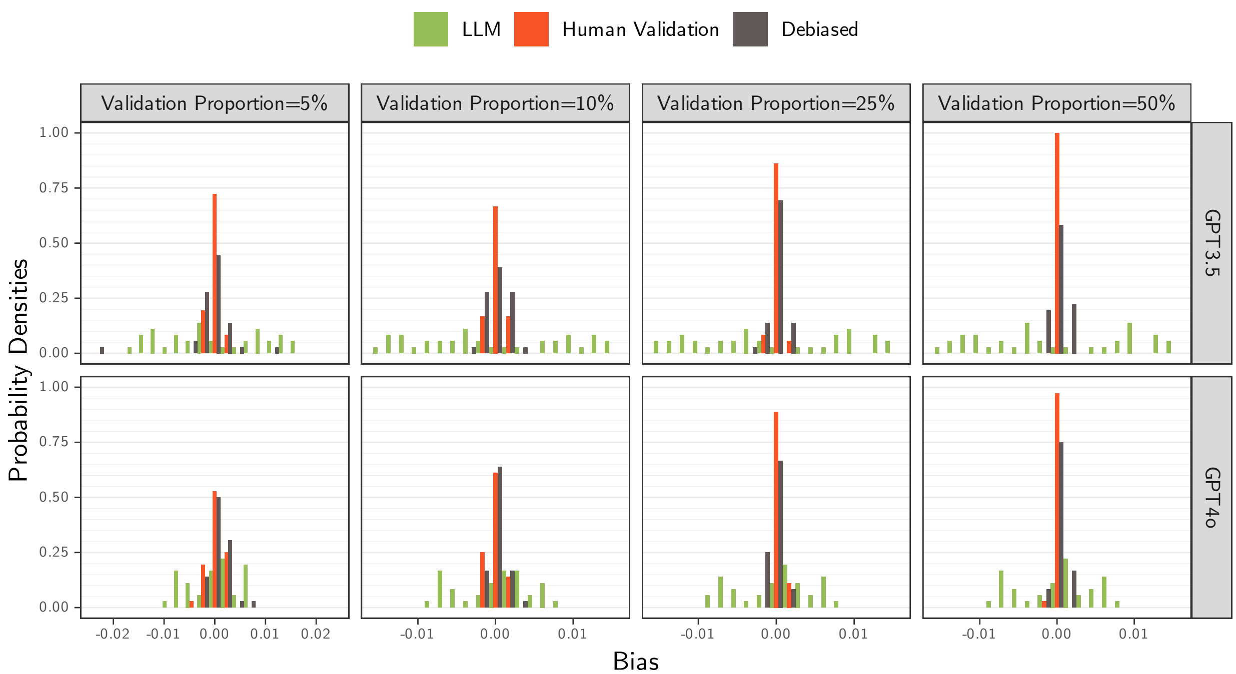


Figure 20: Normalized Bias Density of β_3 by Model and Proportion of Validation Samples. Proxy on the RHS.

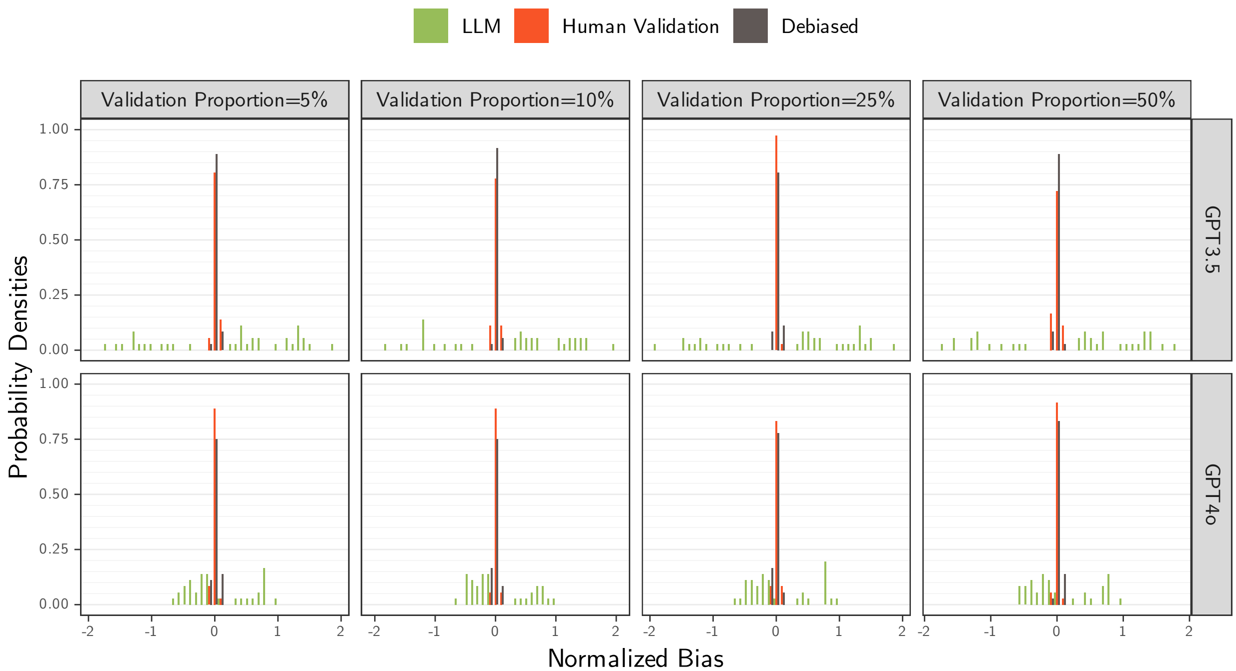


Figure 21: Normalized Bias Density of β_{14} by Model and Proportion of Validation Samples. Proxy on the RHS.

Figure 22: Normalized Bias Density of β_{15} by Model and Proportion of Validation Samples. Proxy on the RHS.

Figure 23: Normalized Bias Density of β_{19} by Model and Proportion of Validation Samples. Proxy on the RHS.

Figure 24: Normalized Bias Density of β_{20} by Model and Proportion of Validation Samples. Proxy on the RHS.

Figure 25: Normalized Bias Density of β_{Other} by Model and Proportion of Validation Samples. Proxy on the RHS.

Figure 26: MSE Empirical CDF for all βs by Proportion of Validation Samples. Proxy on the RHS.

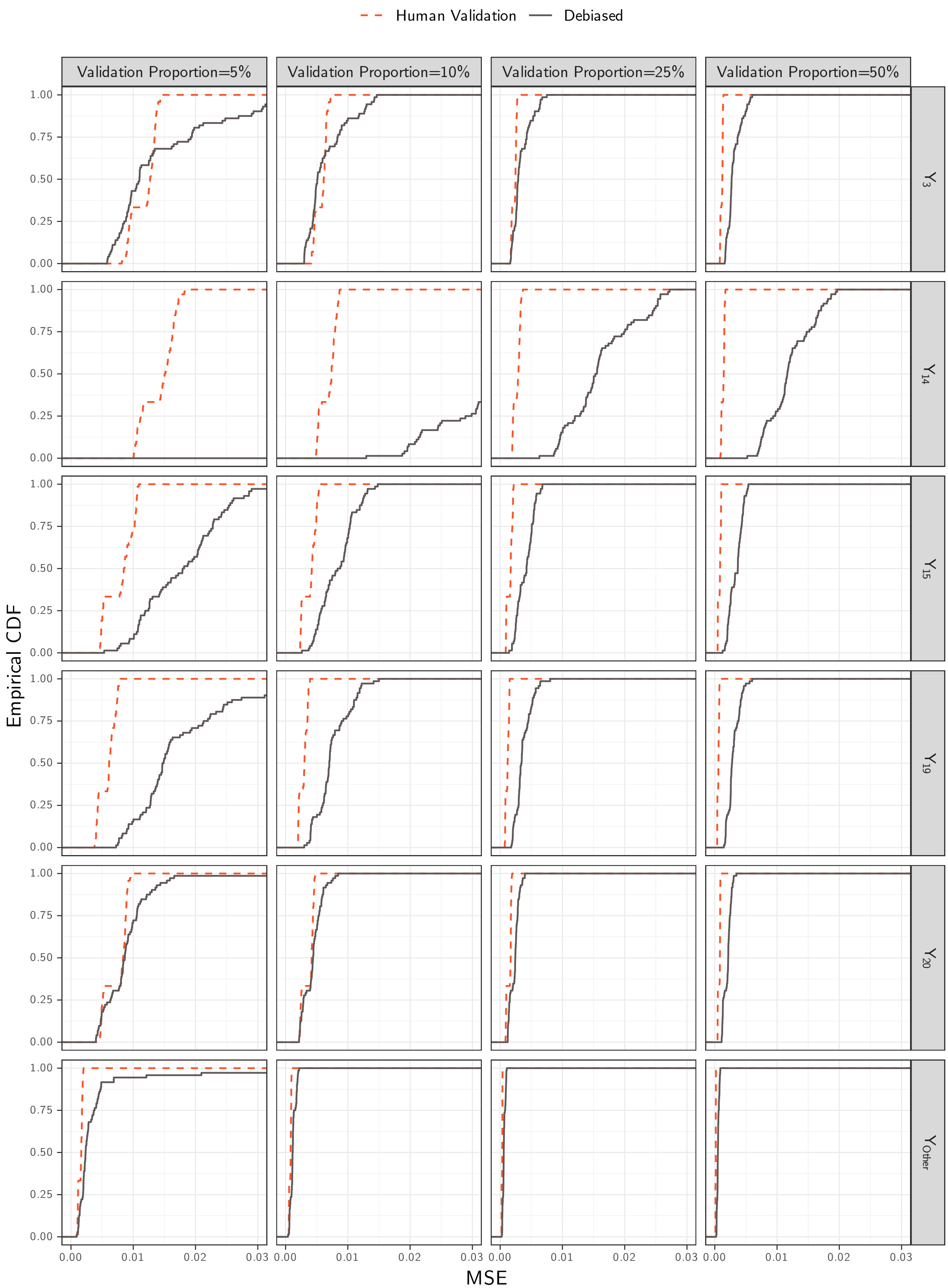


Figure 27: MSE Empirical CDF of β_3 by Model and Proportion of Validation Samples. Proxy on the RHS.

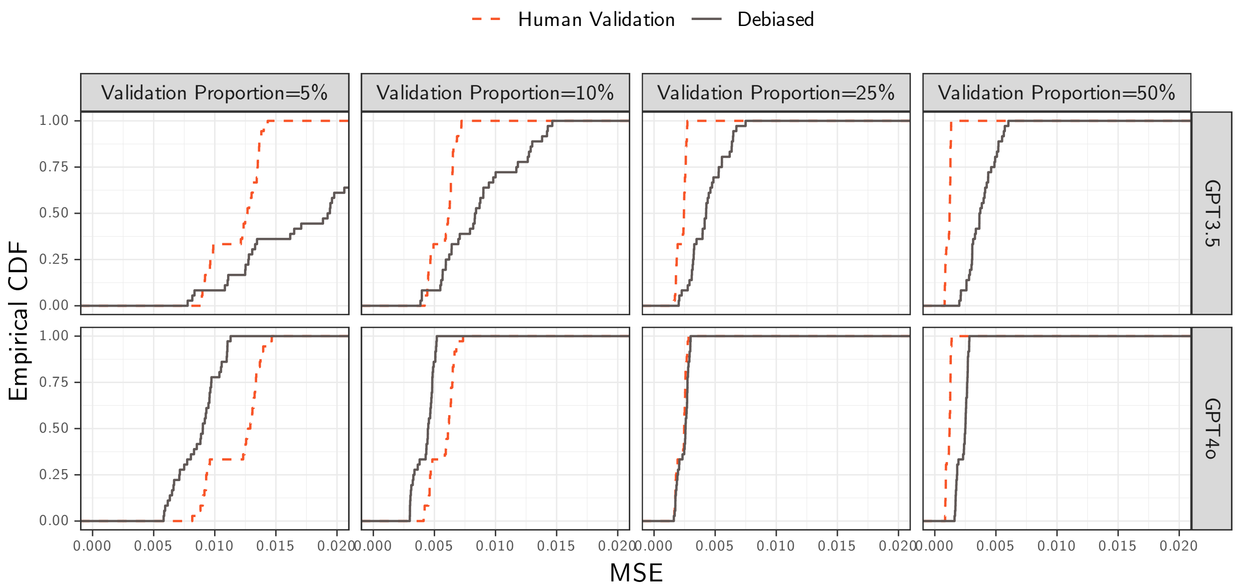


Figure 28: MSE Empirical CDF of β_{14} by Model and Proportion of Validation Samples. Proxy on the RHS.

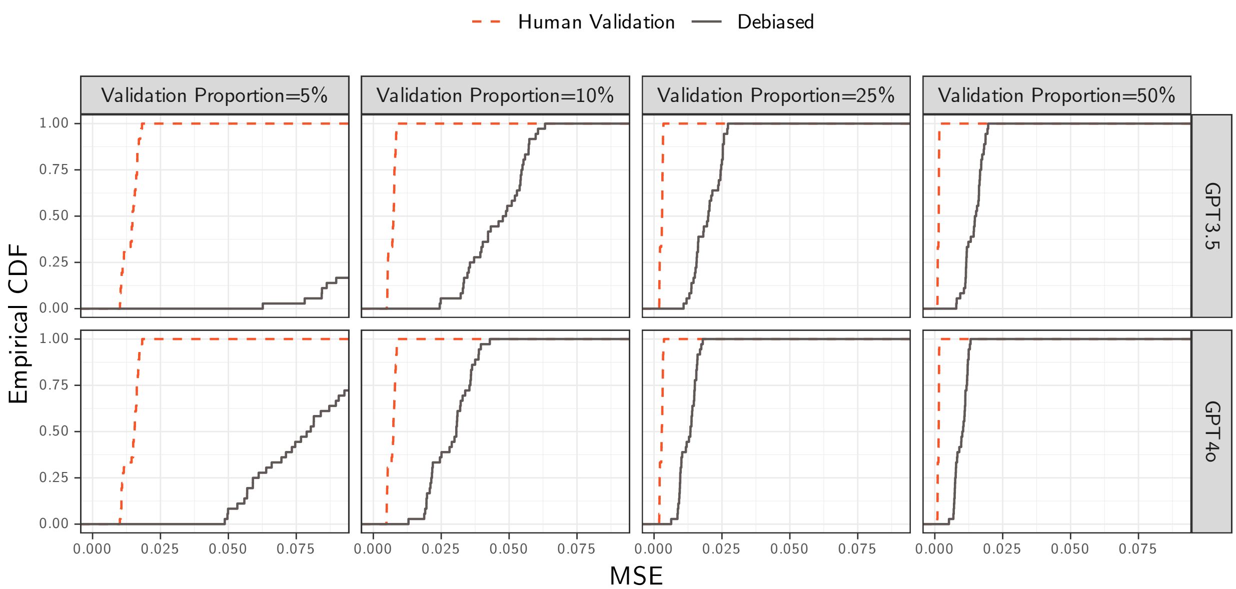


Figure 29: MSE Empirical CDF of β_{15} by Model and Proportion of Validation Samples. Proxy on the RHS.

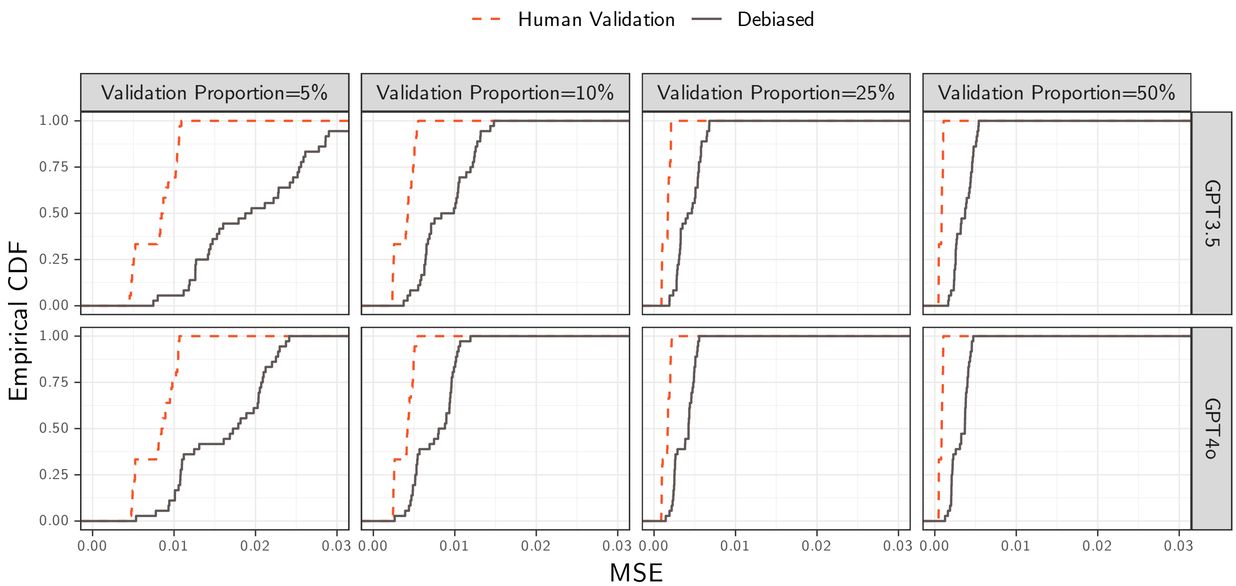


Figure 30: MSE Empirical CDF of β_{19} by Model and Proportion of Validation Samples. Proxy on the RHS.

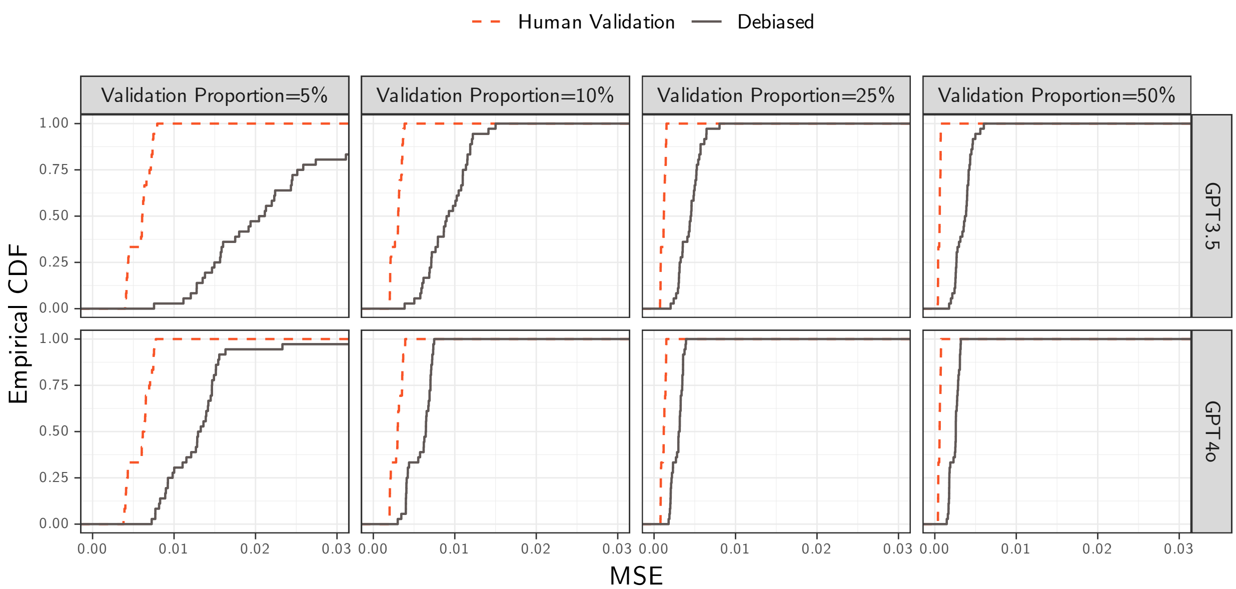


Figure 31: MSE Empirical CDF of β_{20} by Model and Proportion of Validation Samples. Proxy on the RHS.

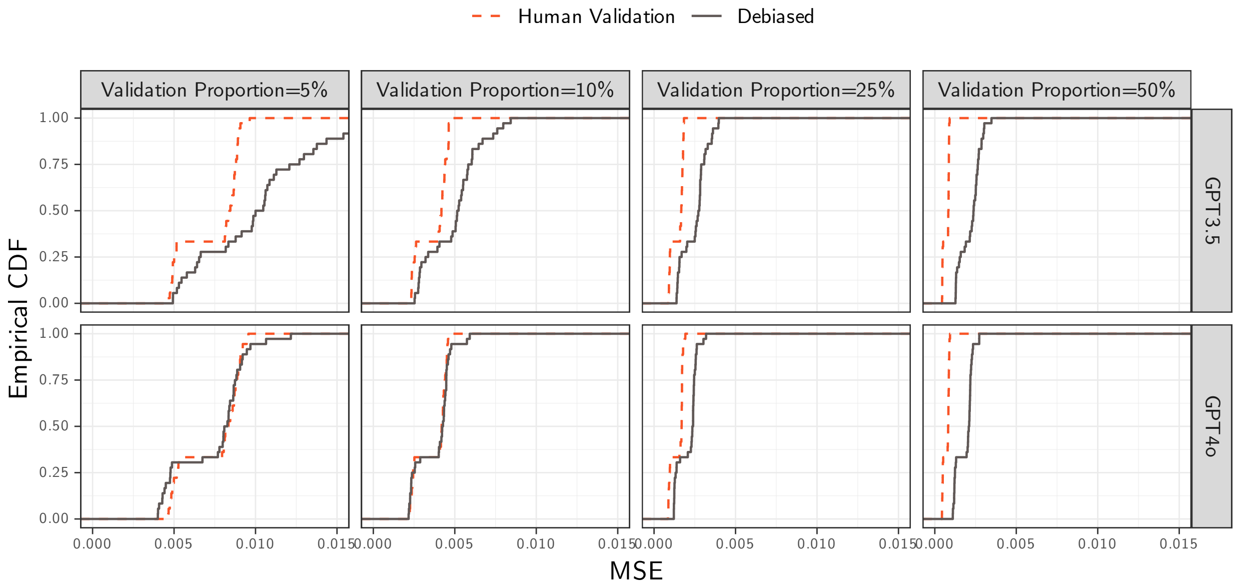


Figure 32: MSE Empirical CDF of β_{Other} by Model and Proportion of Validation Samples. Proxy on the RHS.

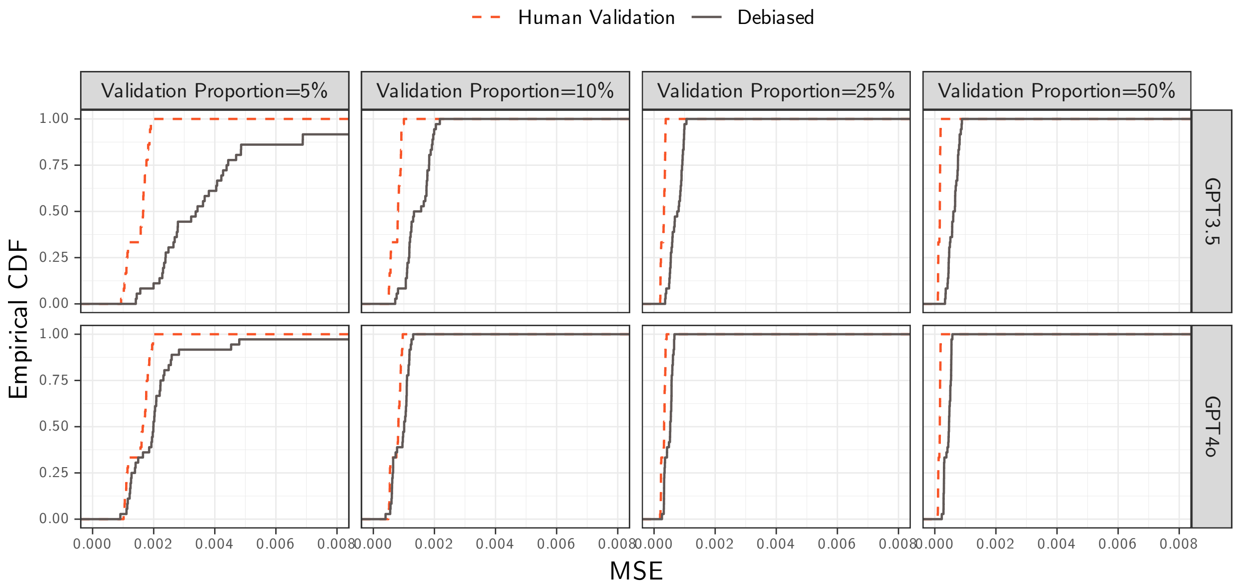


Table 9: Bias Summary Statistics of β_3 by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top\beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	0.003	0.021	0.004	-0.035	0.035
5%	Human Validation	0.000	0.003	0.001	-0.005	0.006
5%	Debiased	0.001	0.004	0.000	-0.005	0.006
10%	LLM	0.003	0.022	0.003	-0.035	0.035
10%	Human Validation	0.000	0.002	0.000	-0.004	0.004
10%	Debiased	0.000	0.003	0.000	-0.004	0.004
25%	LLM	0.003	0.022	0.004	-0.036	0.035
25%	Human Validation	0.000	0.001	0.000	-0.002	0.002
25%	Debiased	0.000	0.002	0.000	-0.003	0.003
50%	LLM	0.003	0.022	0.003	-0.036	0.036
50%	Human Validation	0.000	0.001	0.000	-0.002	0.002
50%	Debiased	0.000	0.002	0.000	-0.002	0.003

Table 10: Bias Summary Statistics of β_{14} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top\beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	-0.003	0.031	-0.012	-0.043	0.049
5%	Human Validation	0.000	0.003	0.000	-0.006	0.005
5%	Debiased	0.007	0.063	0.000	-0.030	0.042
10%	LLM	-0.003	0.031	-0.012	-0.043	0.050
10%	Human Validation	0.000	0.002	-0.001	-0.004	0.003
10%	Debiased	0.001	0.008	0.001	-0.010	0.013
25%	LLM	-0.003	0.031	-0.012	-0.043	0.050
25%	Human Validation	0.000	0.002	0.000	-0.003	0.003
25%	Debiased	0.000	0.004	0.001	-0.007	0.006
50%	LLM	-0.003	0.031	-0.012	-0.043	0.048
50%	Human Validation	0.000	0.001	0.000	-0.001	0.002
50%	Debiased	0.000	0.004	0.000	-0.005	0.006

Table 11: Bias Summary Statistics of β_{15} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top\beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	0.003	0.011	0.003	-0.017	0.020
5%	Human Validation	0.000	0.003	0.000	-0.005	0.004
5%	Debiased	0.000	0.004	0.000	-0.007	0.006
10%	LLM	0.002	0.011	0.003	-0.018	0.020
10%	Human Validation	0.000	0.002	0.000	-0.003	0.003
10%	Debiased	0.000	0.003	0.000	-0.005	0.004
25%	LLM	0.003	0.011	0.003	-0.017	0.020
25%	Human Validation	0.000	0.001	0.000	-0.002	0.002
25%	Debiased	0.000	0.002	0.001	-0.003	0.003
50%	LLM	0.003	0.011	0.003	-0.017	0.019
50%	Human Validation	0.000	0.001	0.000	-0.002	0.001
50%	Debiased	0.000	0.002	0.000	-0.003	0.003

Table 12: Bias Summary Statistics of β_{19} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top\beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	0.007	0.012	0.009	-0.011	0.023
5%	Human Validation	0.001	0.003	0.001	-0.004	0.005
5%	Debiased	-0.002	0.009	-0.001	-0.010	0.005
10%	LLM	0.007	0.012	0.009	-0.011	0.024
10%	Human Validation	0.000	0.002	0.000	-0.003	0.002
10%	Debiased	0.000	0.003	0.000	-0.005	0.004
25%	LLM	0.006	0.012	0.009	-0.011	0.024
25%	Human Validation	0.000	0.001	0.000	-0.002	0.002
25%	Debiased	0.000	0.002	0.000	-0.003	0.003
50%	LLM	0.006	0.012	0.009	-0.011	0.024
50%	Human Validation	0.000	0.001	0.000	-0.001	0.001
50%	Debiased	0.000	0.002	0.000	-0.003	0.003

Table 13: Bias Summary Statistics of β_{20} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top\beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	-0.001	0.013	0.001	-0.022	0.016
5%	Human Validation	0.000	0.003	0.000	-0.005	0.004
5%	Debiased	0.000	0.003	0.000	-0.004	0.005
10%	LLM	-0.001	0.013	0.001	-0.023	0.015
10%	Human Validation	0.000	0.002	0.000	-0.003	0.003
10%	Debiased	0.000	0.002	0.000	-0.003	0.004
25%	LLM	-0.001	0.013	0.001	-0.021	0.015
25%	Human Validation	0.000	0.001	0.000	-0.002	0.002
25%	Debiased	0.000	0.001	0.000	-0.002	0.003
50%	LLM	-0.001	0.012	0.001	-0.021	0.015
50%	Human Validation	0.000	0.001	0.000	-0.001	0.002
50%	Debiased	0.000	0.002	0.000	-0.003	0.003

Table 14: Bias Summary Statistics of β_{Other} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top\beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	0	0.008	0	-0.012	0.013
5%	Human Validation	0	0.001	0	-0.002	0.002
5%	Debiased	0	0.003	0	-0.003	0.003
10%	LLM	0	0.008	0	-0.012	0.013
10%	Human Validation	0	0.001	0	-0.001	0.001
10%	Debiased	0	0.001	0	-0.002	0.002
25%	LLM	0	0.008	0	-0.012	0.014
25%	Human Validation	0	0.001	0	-0.001	0.001
25%	Debiased	0	0.001	0	-0.002	0.001
50%	LLM	0	0.008	0	-0.012	0.013
50%	Human Validation	0	0.000	0	-0.001	0.001
50%	Debiased	0	0.001	0	-0.001	0.001

Table 15: Normalized Bias Summary Statistics of β_3 by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	0.148	0.846	0.166	-1.236	1.398
5%	Human Validation	0.001	0.032	0.005	-0.052	0.057
5%	Debiased	0.003	0.033	0.003	-0.051	0.051
10%	LLM	0.150	0.842	0.139	-1.202	1.419
10%	Human Validation	-0.002	0.029	-0.006	-0.051	0.049
10%	Debiased	0.000	0.033	-0.003	-0.056	0.050
25%	LLM	0.154	0.852	0.165	-1.287	1.419
25%	Human Validation	0.006	0.028	0.008	-0.040	0.047
25%	Debiased	-0.001	0.035	-0.001	-0.055	0.057
50%	LLM	0.150	0.845	0.124	-1.286	1.444
50%	Human Validation	0.001	0.032	0.003	-0.053	0.045
50%	Debiased	-0.002	0.029	-0.004	-0.046	0.047

Table 16: Normalized Bias Summary Statistics of β_{14} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	0.013	1.541	-0.537	-1.885	2.673
5%	Human Validation	-0.002	0.029	-0.001	-0.050	0.044
5%	Debiased	0.004	0.039	0.002	-0.057	0.063
10%	LLM	0.013	1.543	-0.553	-1.929	2.667
10%	Human Validation	-0.006	0.028	-0.006	-0.046	0.043
10%	Debiased	0.002	0.039	0.003	-0.058	0.067
25%	LLM	0.005	1.541	-0.560	-1.934	2.622
25%	Human Validation	0.001	0.032	-0.001	-0.047	0.050
25%	Debiased	-0.001	0.034	0.005	-0.059	0.056
50%	LLM	0.009	1.547	-0.543	-1.968	2.667
50%	Human Validation	0.003	0.032	0.004	-0.042	0.050
50%	Debiased	0.001	0.034	0.001	-0.057	0.057

Table 17: Normalized Bias Summary Statistics of β_{15} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	0.118	0.463	0.166	-0.633	0.791
5%	Human Validation	-0.003	0.033	-0.004	-0.056	0.050
5%	Debiased	-0.002	0.030	-0.002	-0.046	0.045
10%	LLM	0.108	0.470	0.167	-0.695	0.764
10%	Human Validation	-0.001	0.034	0.000	-0.047	0.056
10%	Debiased	-0.003	0.033	-0.002	-0.060	0.048
25%	LLM	0.121	0.460	0.150	-0.653	0.782
25%	Human Validation	-0.005	0.034	-0.012	-0.055	0.056
25%	Debiased	0.004	0.035	0.007	-0.052	0.054
50%	LLM	0.112	0.450	0.158	-0.649	0.762
50%	Human Validation	-0.006	0.028	-0.003	-0.052	0.035
50%	Debiased	-0.002	0.030	0.000	-0.051	0.042

Table 18: Normalized Bias Summary Statistics of β_{19} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	0.369	0.615	0.486	-0.534	1.285
5%	Human Validation	0.008	0.034	0.010	-0.050	0.062
5%	Debiased	-0.010	0.034	-0.009	-0.064	0.039
10%	LLM	0.368	0.615	0.466	-0.502	1.344
10%	Human Validation	-0.004	0.029	-0.006	-0.044	0.039
10%	Debiased	-0.004	0.034	-0.003	-0.064	0.048
25%	LLM	0.359	0.607	0.451	-0.493	1.326
25%	Human Validation	-0.001	0.032	-0.001	-0.058	0.055
25%	Debiased	0.000	0.033	-0.002	-0.045	0.054
50%	LLM	0.361	0.619	0.466	-0.554	1.331
50%	Human Validation	0.007	0.027	0.007	-0.034	0.047
50%	Debiased	-0.002	0.035	0.000	-0.069	0.049

Table 19: Normalized Bias Summary Statistics of β_{20} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	-0.008	0.632	0.023	-1.000	0.888
5%	Human Validation	-0.001	0.034	0.001	-0.059	0.047
5%	Debiased	0.002	0.030	0.001	-0.041	0.048
10%	LLM	-0.002	0.630	0.041	-1.005	0.864
10%	Human Validation	0.001	0.030	0.002	-0.049	0.043
10%	Debiased	0.004	0.033	0.004	-0.048	0.056
25%	LLM	-0.009	0.623	0.029	-0.970	0.838
25%	Human Validation	0.003	0.030	0.004	-0.049	0.047
25%	Debiased	-0.001	0.030	-0.007	-0.040	0.060
50%	LLM	-0.003	0.619	0.049	-0.990	0.835
50%	Human Validation	0.001	0.034	0.000	-0.055	0.054
50%	Debiased	0.000	0.036	0.003	-0.055	0.061

Table 20: Normalized Bias Summary Statistics of β_{Other} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	-0.093	0.979	0.030	-1.766	1.413
5%	Human Validation	-0.003	0.033	-0.006	-0.051	0.055
5%	Debiased	0.001	0.031	-0.006	-0.046	0.055
10%	LLM	-0.100	0.961	0.021	-1.788	1.394
10%	Human Validation	-0.001	0.030	0.001	-0.049	0.044
10%	Debiased	0.000	0.035	-0.004	-0.049	0.060
25%	LLM	-0.100	0.970	0.016	-1.767	1.439
25%	Human Validation	0.001	0.030	-0.002	-0.045	0.053
25%	Debiased	-0.005	0.036	-0.004	-0.065	0.055
50%	LLM	-0.082	0.957	0.036	-1.721	1.424
50%	Human Validation	-0.001	0.034	0.008	-0.057	0.043
50%	Debiased	0.004	0.037	0.001	-0.050	0.060

Table 21: Bias Summary Statistics of β_3 by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.004	0.028	0.013	-0.042	0.037
GPT-3.5	5%	Human Validation	0.001	0.004	0.001	-0.005	0.006
GPT-3.5	5%	Debiased	0.001	0.005	0.000	-0.005	0.007
GPT-3.5	10%	LLM	0.004	0.029	0.012	-0.041	0.038
GPT-3.5	10%	Human Validation	0.000	0.002	0.000	-0.004	0.004
GPT-3.5	10%	Debiased	0.000	0.003	0.000	-0.003	0.004
GPT-3.5	25%	LLM	0.004	0.029	0.012	-0.041	0.038
GPT-3.5	25%	Human Validation	0.000	0.001	0.000	-0.002	0.002
GPT-3.5	25%	Debiased	0.000	0.002	0.000	-0.003	0.004
GPT-3.5	50%	LLM	0.004	0.029	0.012	-0.041	0.037
GPT-3.5	50%	Human Validation	0.000	0.001	0.000	-0.002	0.002
GPT-3.5	50%	Debiased	0.000	0.002	0.000	-0.003	0.003
GPT-4o	5%	LLM	0.001	0.011	-0.003	-0.012	0.017
GPT-4o	5%	Human Validation	-0.001	0.003	0.000	-0.005	0.004
GPT-4o	5%	Debiased	0.000	0.003	0.000	-0.005	0.005
GPT-4o	10%	LLM	0.001	0.011	-0.003	-0.012	0.017
GPT-4o	10%	Human Validation	0.000	0.002	0.000	-0.003	0.003
GPT-4o	10%	Debiased	0.000	0.002	0.000	-0.004	0.004
GPT-4o	25%	LLM	0.001	0.011	-0.003	-0.012	0.017
GPT-4o	25%	Human Validation	0.000	0.001	0.000	-0.002	0.003
GPT-4o	25%	Debiased	0.000	0.002	0.000	-0.003	0.003
GPT-4o	50%	LLM	0.001	0.011	-0.003	-0.012	0.016
GPT-4o	50%	Human Validation	0.000	0.001	0.000	-0.001	0.001
GPT-4o	50%	Debiased	0.000	0.002	0.000	-0.002	0.003

Table 22: Bias Summary Statistics of β_{14} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	-0.004	0.036	-0.018	-0.044	0.051
GPT-3.5	5%	Human Validation	-0.001	0.004	-0.001	-0.006	0.004
GPT-3.5	5%	Debiased	0.012	0.086	0.000	-0.034	0.051
GPT-3.5	10%	LLM	-0.003	0.036	-0.019	-0.045	0.052
GPT-3.5	10%	Human Validation	-0.001	0.002	-0.001	-0.004	0.003
GPT-3.5	10%	Debiased	0.001	0.009	0.002	-0.012	0.013
GPT-3.5	25%	LLM	-0.004	0.036	-0.017	-0.046	0.052
GPT-3.5	25%	Human Validation	0.000	0.002	0.000	-0.002	0.002
GPT-3.5	25%	Debiased	0.000	0.005	0.001	-0.008	0.006
GPT-3.5	50%	LLM	-0.004	0.036	-0.020	-0.045	0.051
GPT-3.5	50%	Human Validation	0.000	0.001	0.000	-0.001	0.002
GPT-3.5	50%	Debiased	0.001	0.004	0.000	-0.005	0.006
GPT-4o	5%	LLM	-0.002	0.025	-0.008	-0.031	0.036
GPT-4o	5%	Human Validation	0.000	0.003	0.000	-0.005	0.005
GPT-4o	5%	Debiased	0.002	0.021	0.001	-0.021	0.032
GPT-4o	10%	LLM	-0.002	0.025	-0.009	-0.032	0.036
GPT-4o	10%	Human Validation	0.000	0.002	0.000	-0.004	0.003
GPT-4o	10%	Debiased	0.000	0.006	0.000	-0.009	0.012
GPT-4o	25%	LLM	-0.002	0.025	-0.008	-0.032	0.036
GPT-4o	25%	Human Validation	0.000	0.002	0.000	-0.003	0.003
GPT-4o	25%	Debiased	0.000	0.004	0.000	-0.005	0.006
GPT-4o	50%	LLM	-0.002	0.025	-0.009	-0.032	0.035
GPT-4o	50%	Human Validation	0.000	0.001	0.000	-0.002	0.002
GPT-4o	50%	Debiased	0.000	0.004	0.000	-0.005	0.005

Table 23: Bias Summary Statistics of β_{15} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.002	0.013	0.003	-0.018	0.023
GPT-3.5	5%	Human Validation	0.000	0.003	-0.001	-0.004	0.005
GPT-3.5	5%	Debiased	0.000	0.005	0.001	-0.006	0.007
GPT-3.5	10%	LLM	0.001	0.014	0.003	-0.018	0.022
GPT-3.5	10%	Human Validation	0.000	0.002	0.000	-0.003	0.003
GPT-3.5	10%	Debiased	0.000	0.003	0.000	-0.005	0.004
GPT-3.5	25%	LLM	0.002	0.014	0.003	-0.017	0.022
GPT-3.5	25%	Human Validation	0.000	0.001	-0.001	-0.002	0.002
GPT-3.5	25%	Debiased	0.000	0.003	0.000	-0.003	0.004
GPT-3.5	50%	LLM	0.001	0.013	0.003	-0.017	0.022
GPT-3.5	50%	Human Validation	0.000	0.001	0.000	-0.002	0.001
GPT-3.5	50%	Debiased	0.000	0.002	0.000	-0.003	0.003
GPT-4o	5%	LLM	0.004	0.008	0.003	-0.008	0.016
GPT-4o	5%	Human Validation	-0.001	0.003	0.000	-0.005	0.004
GPT-4o	5%	Debiased	-0.001	0.004	-0.001	-0.007	0.004
GPT-4o	10%	LLM	0.003	0.008	0.004	-0.008	0.016
GPT-4o	10%	Human Validation	0.000	0.002	0.000	-0.003	0.004
GPT-4o	10%	Debiased	-0.001	0.003	0.000	-0.004	0.003
GPT-4o	25%	LLM	0.004	0.008	0.003	-0.007	0.017
GPT-4o	25%	Human Validation	0.000	0.001	0.000	-0.002	0.003
GPT-4o	25%	Debiased	0.000	0.002	0.001	-0.004	0.003
GPT-4o	50%	LLM	0.004	0.008	0.003	-0.008	0.017
GPT-4o	50%	Human Validation	0.000	0.001	0.000	-0.002	0.001
GPT-4o	50%	Debiased	0.000	0.002	0.000	-0.003	0.002

Table 24: Bias Summary Statistics of β_{19} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.008	0.015	0.013	-0.013	0.024
GPT-3.5	5%	Human Validation	0.000	0.003	0.000	-0.005	0.005
GPT-3.5	5%	Debiased	-0.003	0.012	-0.001	-0.012	0.005
GPT-3.5	10%	LLM	0.009	0.015	0.014	-0.013	0.024
GPT-3.5	10%	Human Validation	0.000	0.002	0.000	-0.003	0.002
GPT-3.5	10%	Debiased	-0.001	0.003	-0.001	-0.006	0.004
GPT-3.5	25%	LLM	0.008	0.015	0.014	-0.013	0.025
GPT-3.5	25%	Human Validation	0.000	0.001	0.000	-0.001	0.002
GPT-3.5	25%	Debiased	0.000	0.002	0.000	-0.004	0.003
GPT-3.5	50%	LLM	0.008	0.015	0.013	-0.013	0.025
GPT-3.5	50%	Human Validation	0.000	0.001	0.000	-0.001	0.001
GPT-3.5	50%	Debiased	0.000	0.002	0.000	-0.004	0.003
GPT-4o	5%	LLM	0.005	0.008	0.007	-0.008	0.016
GPT-4o	5%	Human Validation	0.001	0.002	0.001	-0.003	0.005
GPT-4o	5%	Debiased	-0.001	0.004	-0.001	-0.007	0.005
GPT-4o	10%	LLM	0.005	0.008	0.007	-0.008	0.017
GPT-4o	10%	Human Validation	0.000	0.002	0.000	-0.003	0.002
GPT-4o	10%	Debiased	0.000	0.003	0.001	-0.005	0.004
GPT-4o	25%	LLM	0.004	0.008	0.006	-0.008	0.015
GPT-4o	25%	Human Validation	0.000	0.001	0.000	-0.002	0.002
GPT-4o	25%	Debiased	0.000	0.002	0.000	-0.002	0.003
GPT-4o	50%	LLM	0.004	0.008	0.006	-0.009	0.016
GPT-4o	50%	Human Validation	0.000	0.001	0.000	-0.001	0.001
GPT-4o	50%	Debiased	0.000	0.002	0.000	-0.003	0.002

Table 25: Bias Summary Statistics of β_{20} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.001	0.011	0.001	-0.015	0.018
GPT-3.5	5%	Human Validation	0.000	0.003	0.000	-0.006	0.005
GPT-3.5	5%	Debiased	0.001	0.003	0.001	-0.004	0.007
GPT-3.5	10%	LLM	0.001	0.011	0.001	-0.015	0.017
GPT-3.5	10%	Human Validation	0.000	0.002	0.000	-0.002	0.003
GPT-3.5	10%	Debiased	0.000	0.003	0.000	-0.004	0.004
GPT-3.5	25%	LLM	0.001	0.011	0.000	-0.015	0.017
GPT-3.5	25%	Human Validation	0.000	0.001	0.000	-0.001	0.002
GPT-3.5	25%	Debiased	0.000	0.002	0.000	-0.002	0.003
GPT-3.5	50%	LLM	0.001	0.011	0.001	-0.015	0.017
GPT-3.5	50%	Human Validation	0.000	0.001	0.000	-0.001	0.002
GPT-3.5	50%	Debiased	0.000	0.002	0.000	-0.003	0.003
GPT-4o	5%	LLM	-0.003	0.014	0.001	-0.022	0.012
GPT-4o	5%	Human Validation	0.000	0.003	0.000	-0.004	0.004
GPT-4o	5%	Debiased	-0.001	0.002	-0.001	-0.004	0.003
GPT-4o	10%	LLM	-0.003	0.014	0.001	-0.023	0.013
GPT-4o	10%	Human Validation	0.000	0.002	0.000	-0.003	0.002
GPT-4o	10%	Debiased	0.000	0.002	0.001	-0.003	0.003
GPT-4o	25%	LLM	-0.003	0.014	0.001	-0.022	0.012
GPT-4o	25%	Human Validation	0.000	0.001	0.000	-0.002	0.002
GPT-4o	25%	Debiased	0.000	0.001	0.000	-0.002	0.001
GPT-4o	50%	LLM	-0.003	0.014	0.001	-0.022	0.012
GPT-4o	50%	Human Validation	0.000	0.001	0.000	-0.002	0.002
GPT-4o	50%	Debiased	0.000	0.001	0.000	-0.002	0.002

Table 26: Bias Summary Statistics of β_{Other} by Model and Proportion of Validation Samples for. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.000	0.010	-0.002	-0.013	0.014
GPT-3.5	5%	Human Validation	0.000	0.001	0.000	-0.002	0.002
GPT-3.5	5%	Debiased	-0.001	0.004	-0.001	-0.004	0.003
GPT-3.5	10%	LLM	0.000	0.010	-0.003	-0.013	0.014
GPT-3.5	10%	Human Validation	0.000	0.001	0.000	-0.001	0.001
GPT-3.5	10%	Debiased	0.000	0.001	0.000	-0.002	0.002
GPT-3.5	25%	LLM	0.000	0.010	-0.002	-0.013	0.014
GPT-3.5	25%	Human Validation	0.000	0.001	0.000	-0.001	0.001
GPT-3.5	25%	Debiased	0.000	0.001	0.000	-0.002	0.001
GPT-3.5	50%	LLM	0.000	0.010	-0.003	-0.013	0.014
GPT-3.5	50%	Human Validation	0.000	0.000	0.000	-0.001	0.001
GPT-3.5	50%	Debiased	0.000	0.001	0.000	-0.001	0.001
GPT-4o	5%	LLM	0.000	0.005	0.001	-0.007	0.007
GPT-4o	5%	Human Validation	0.000	0.001	0.000	-0.002	0.002
GPT-4o	5%	Debiased	0.000	0.002	0.000	-0.002	0.003
GPT-4o	10%	LLM	0.000	0.005	0.001	-0.007	0.007
GPT-4o	10%	Human Validation	0.000	0.001	0.000	-0.001	0.001
GPT-4o	10%	Debiased	0.000	0.001	0.000	-0.002	0.002
GPT-4o	25%	LLM	0.000	0.005	0.001	-0.007	0.007
GPT-4o	25%	Human Validation	0.000	0.001	0.000	-0.001	0.001
GPT-4o	25%	Debiased	0.000	0.001	0.000	-0.001	0.001
GPT-4o	50%	LLM	0.000	0.005	0.001	-0.007	0.007
GPT-4o	50%	Human Validation	0.000	0.000	0.000	-0.001	0.001
GPT-4o	50%	Debiased	0.000	0.001	0.000	-0.001	0.001

Table 27: Normalized Bias Summary of β_3 Statistics by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.212	1.089	0.483	-1.480	1.463
GPT-3.5	5%	Human Validation	0.008	0.034	0.006	-0.049	0.063
GPT-3.5	5%	Debiased	0.001	0.028	0.001	-0.042	0.048
GPT-3.5	10%	LLM	0.225	1.085	0.500	-1.453	1.500
GPT-3.5	10%	Human Validation	0.000	0.030	-0.005	-0.051	0.055
GPT-3.5	10%	Debiased	0.005	0.028	0.006	-0.032	0.046
GPT-3.5	25%	LLM	0.230	1.094	0.538	-1.462	1.493
GPT-3.5	25%	Human Validation	0.006	0.025	0.011	-0.031	0.041
GPT-3.5	25%	Debiased	0.002	0.034	0.001	-0.045	0.057
GPT-3.5	50%	LLM	0.222	1.091	0.515	-1.490	1.519
GPT-3.5	50%	Human Validation	-0.001	0.034	-0.005	-0.055	0.051
GPT-3.5	50%	Debiased	-0.006	0.027	-0.004	-0.050	0.038
GPT-4o	5%	LLM	0.084	0.507	-0.119	-0.510	0.824
GPT-4o	5%	Human Validation	-0.005	0.029	-0.003	-0.051	0.037
GPT-4o	5%	Debiased	0.005	0.037	0.005	-0.067	0.055
GPT-4o	10%	LLM	0.075	0.501	-0.135	-0.468	0.825
GPT-4o	10%	Human Validation	-0.005	0.028	-0.007	-0.043	0.042
GPT-4o	10%	Debiased	-0.006	0.036	-0.006	-0.074	0.052
GPT-4o	25%	LLM	0.078	0.513	-0.136	-0.483	0.857
GPT-4o	25%	Human Validation	0.006	0.032	0.006	-0.048	0.055
GPT-4o	25%	Debiased	-0.005	0.036	-0.001	-0.061	0.048
GPT-4o	50%	LLM	0.078	0.497	-0.119	-0.514	0.805
GPT-4o	50%	Human Validation	0.003	0.029	0.013	-0.041	0.038
GPT-4o	50%	Debiased	0.001	0.032	-0.004	-0.041	0.066

Table 28: Normalized Bias Summary Statistics of β_{14} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	-0.003	1.821	-0.848	-2.053	2.819
GPT-3.5	5%	Human Validation	-0.006	0.030	-0.006	-0.053	0.032
GPT-3.5	5%	Debiased	0.003	0.035	0.000	-0.057	0.061
GPT-3.5	10%	LLM	0.003	1.826	-0.853	-2.059	2.826
GPT-3.5	10%	Human Validation	-0.012	0.028	-0.012	-0.048	0.039
GPT-3.5	10%	Debiased	0.002	0.043	0.009	-0.062	0.061
GPT-3.5	25%	LLM	-0.021	1.822	-0.808	-2.035	2.855
GPT-3.5	25%	Human Validation	0.002	0.029	0.002	-0.035	0.044
GPT-3.5	25%	Debiased	-0.001	0.034	0.006	-0.068	0.039
GPT-3.5	50%	LLM	-0.003	1.832	-0.886	-2.059	2.815
GPT-3.5	50%	Human Validation	0.012	0.031	0.011	-0.039	0.053
GPT-3.5	50%	Debiased	0.005	0.031	0.001	-0.042	0.053
GPT-4o	5%	LLM	0.028	1.224	-0.369	-1.314	1.908
GPT-4o	5%	Human Validation	0.003	0.028	0.004	-0.043	0.045
GPT-4o	5%	Debiased	0.006	0.043	0.004	-0.056	0.067
GPT-4o	10%	LLM	0.023	1.224	-0.370	-1.344	1.890
GPT-4o	10%	Human Validation	0.000	0.027	0.001	-0.040	0.044
GPT-4o	10%	Debiased	0.001	0.035	0.002	-0.046	0.073
GPT-4o	25%	LLM	0.031	1.222	-0.332	-1.323	1.879
GPT-4o	25%	Human Validation	0.000	0.036	-0.003	-0.057	0.053
GPT-4o	25%	Debiased	0.000	0.033	0.000	-0.043	0.056
GPT-4o	50%	LLM	0.022	1.223	-0.414	-1.309	1.859
GPT-4o	50%	Human Validation	-0.006	0.030	-0.010	-0.043	0.047
GPT-4o	50%	Debiased	-0.004	0.038	0.000	-0.061	0.055

Table 29: Normalized Bias Summary Statistics of β_{15} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.075	0.568	0.189	-0.712	0.900
GPT-3.5	5%	Human Validation	0.000	0.033	-0.007	-0.052	0.056
GPT-3.5	5%	Debiased	0.002	0.032	0.004	-0.039	0.049
GPT-3.5	10%	LLM	0.069	0.577	0.167	-0.725	0.950
GPT-3.5	10%	Human Validation	-0.002	0.035	0.002	-0.041	0.047
GPT-3.5	10%	Debiased	-0.001	0.035	-0.002	-0.062	0.048
GPT-3.5	25%	LLM	0.073	0.566	0.150	-0.688	0.930
GPT-3.5	25%	Human Validation	-0.011	0.031	-0.015	-0.056	0.043
GPT-3.5	25%	Debiased	0.005	0.039	0.005	-0.046	0.061
GPT-3.5	50%	LLM	0.072	0.550	0.132	-0.683	0.902
GPT-3.5	50%	Human Validation	-0.007	0.025	-0.002	-0.048	0.032
GPT-3.5	50%	Debiased	-0.004	0.031	-0.007	-0.051	0.045
GPT-4o	5%	LLM	0.160	0.328	0.140	-0.276	0.693
GPT-4o	5%	Human Validation	-0.006	0.033	0.000	-0.055	0.048
GPT-4o	5%	Debiased	-0.006	0.028	-0.006	-0.050	0.033
GPT-4o	10%	LLM	0.148	0.334	0.171	-0.283	0.664
GPT-4o	10%	Human Validation	0.001	0.033	-0.003	-0.049	0.064
GPT-4o	10%	Debiased	-0.005	0.030	-0.002	-0.055	0.037
GPT-4o	25%	LLM	0.169	0.322	0.152	-0.241	0.685
GPT-4o	25%	Human Validation	0.001	0.037	-0.006	-0.045	0.071
GPT-4o	25%	Debiased	0.004	0.031	0.010	-0.056	0.044
GPT-4o	50%	LLM	0.152	0.324	0.158	-0.300	0.654
GPT-4o	50%	Human Validation	-0.005	0.032	-0.005	-0.055	0.038
GPT-4o	50%	Debiased	0.000	0.030	0.003	-0.054	0.040

Table 30: Normalized Bias Summary Statistics of β_{19} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.448	0.723	0.594	-0.586	1.374
GPT-3.5	5%	Human Validation	0.004	0.038	0.004	-0.059	0.064
GPT-3.5	5%	Debiased	-0.009	0.035	-0.008	-0.062	0.033
GPT-3.5	10%	LLM	0.452	0.728	0.665	-0.575	1.387
GPT-3.5	10%	Human Validation	-0.003	0.030	-0.007	-0.046	0.034
GPT-3.5	10%	Debiased	-0.007	0.034	-0.008	-0.061	0.048
GPT-3.5	25%	LLM	0.443	0.723	0.642	-0.572	1.402
GPT-3.5	25%	Human Validation	-0.001	0.030	-0.002	-0.043	0.062
GPT-3.5	25%	Debiased	-0.002	0.034	-0.004	-0.051	0.046
GPT-3.5	50%	LLM	0.447	0.737	0.622	-0.620	1.413
GPT-3.5	50%	Human Validation	0.008	0.026	0.008	-0.032	0.049
GPT-3.5	50%	Debiased	0.002	0.036	0.005	-0.058	0.052
GPT-4o	5%	LLM	0.290	0.482	0.423	-0.435	0.905
GPT-4o	5%	Human Validation	0.011	0.030	0.015	-0.043	0.054
GPT-4o	5%	Debiased	-0.010	0.033	-0.011	-0.064	0.047
GPT-4o	10%	LLM	0.285	0.471	0.412	-0.420	0.914
GPT-4o	10%	Human Validation	-0.004	0.028	-0.006	-0.043	0.041
GPT-4o	10%	Debiased	-0.001	0.035	0.010	-0.064	0.046
GPT-4o	25%	LLM	0.276	0.458	0.392	-0.441	0.836
GPT-4o	25%	Human Validation	-0.001	0.033	0.000	-0.061	0.050
GPT-4o	25%	Debiased	0.003	0.032	0.000	-0.035	0.060
GPT-4o	50%	LLM	0.275	0.468	0.382	-0.447	0.859
GPT-4o	50%	Human Validation	0.005	0.029	0.005	-0.034	0.041
GPT-4o	50%	Debiased	-0.006	0.033	-0.005	-0.068	0.041

Table 31: Normalized Bias Summary Statistics of β_{20} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.068	0.557	0.023	-0.676	0.977
GPT-3.5	5%	Human Validation	0.001	0.038	0.003	-0.070	0.052
GPT-3.5	5%	Debiased	0.009	0.030	0.012	-0.040	0.053
GPT-3.5	10%	LLM	0.072	0.558	0.040	-0.657	0.938
GPT-3.5	10%	Human Validation	0.005	0.032	0.001	-0.038	0.055
GPT-3.5	10%	Debiased	0.000	0.036	0.001	-0.053	0.059
GPT-3.5	25%	LLM	0.070	0.561	0.012	-0.680	0.950
GPT-3.5	25%	Human Validation	0.012	0.028	0.011	-0.031	0.057
GPT-3.5	25%	Debiased	0.002	0.032	-0.008	-0.037	0.066
GPT-3.5	50%	LLM	0.076	0.554	0.052	-0.673	0.943
GPT-3.5	50%	Human Validation	0.004	0.029	-0.002	-0.037	0.053
GPT-3.5	50%	Debiased	-0.001	0.038	0.003	-0.059	0.059
GPT-4o	5%	LLM	-0.085	0.697	0.038	-1.051	0.757
GPT-4o	5%	Human Validation	-0.003	0.031	0.001	-0.050	0.042
GPT-4o	5%	Debiased	-0.006	0.028	-0.010	-0.039	0.041
GPT-4o	10%	LLM	-0.076	0.695	0.041	-1.062	0.803
GPT-4o	10%	Human Validation	-0.003	0.028	0.002	-0.052	0.032
GPT-4o	10%	Debiased	0.008	0.029	0.010	-0.041	0.050
GPT-4o	25%	LLM	-0.087	0.677	0.042	-1.030	0.752
GPT-4o	25%	Human Validation	-0.005	0.031	0.001	-0.057	0.044
GPT-4o	25%	Debiased	-0.003	0.029	-0.007	-0.044	0.041
GPT-4o	50%	LLM	-0.083	0.677	0.047	-1.015	0.699
GPT-4o	50%	Human Validation	-0.001	0.038	0.003	-0.068	0.053
GPT-4o	50%	Debiased	0.001	0.033	0.001	-0.045	0.059

Table 32: Normalized Bias Summary Statistics of β_{Other} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	-0.145	1.247	-0.255	-1.990	1.618
GPT-3.5	5%	Human Validation	-0.009	0.028	-0.010	-0.055	0.040
GPT-3.5	5%	Debiased	-0.006	0.028	-0.012	-0.045	0.033
GPT-3.5	10%	LLM	-0.161	1.223	-0.290	-1.927	1.580
GPT-3.5	10%	Human Validation	0.001	0.031	0.004	-0.048	0.045
GPT-3.5	10%	Debiased	0.000	0.037	-0.001	-0.048	0.052
GPT-3.5	25%	LLM	-0.154	1.227	-0.290	-1.958	1.584
GPT-3.5	25%	Human Validation	-0.004	0.030	-0.003	-0.059	0.043
GPT-3.5	25%	Debiased	-0.003	0.036	-0.002	-0.065	0.054
GPT-3.5	50%	LLM	-0.137	1.218	-0.313	-1.874	1.525
GPT-3.5	50%	Human Validation	0.000	0.034	0.006	-0.056	0.043
GPT-3.5	50%	Debiased	0.001	0.037	-0.003	-0.050	0.055
GPT-4o	5%	LLM	-0.041	0.620	0.099	-1.010	0.731
GPT-4o	5%	Human Validation	0.002	0.036	0.005	-0.047	0.060
GPT-4o	5%	Debiased	0.009	0.034	0.007	-0.039	0.057
GPT-4o	10%	LLM	-0.039	0.608	0.119	-0.984	0.722
GPT-4o	10%	Human Validation	-0.004	0.029	0.000	-0.042	0.041
GPT-4o	10%	Debiased	0.001	0.033	-0.005	-0.049	0.060
GPT-4o	25%	LLM	-0.047	0.630	0.100	-1.064	0.736
GPT-4o	25%	Human Validation	0.007	0.030	0.003	-0.033	0.055
GPT-4o	25%	Debiased	-0.008	0.037	-0.007	-0.060	0.052
GPT-4o	50%	LLM	-0.026	0.607	0.120	-0.998	0.739
GPT-4o	50%	Human Validation	-0.002	0.035	0.008	-0.056	0.044
GPT-4o	50%	Debiased	0.008	0.037	0.005	-0.047	0.069

Table 33: MSE Summary Statistics of β_3 by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.001	0.001	0.001	0.001	0.002
5%	Human Validation	0.012	0.002	0.013	0.009	0.014
5%	Debiased	0.015	0.015	0.011	0.006	0.031
10%	LLM	0.001	0.001	0.001	0.001	0.002
10%	Human Validation	0.006	0.001	0.006	0.004	0.007
10%	Debiased	0.007	0.003	0.005	0.003	0.013
25%	LLM	0.001	0.001	0.001	0.001	0.003
25%	Human Validation	0.002	0.000	0.002	0.002	0.003
25%	Debiased	0.003	0.001	0.003	0.002	0.006
50%	LLM	0.001	0.001	0.001	0.001	0.002
50%	Human Validation	0.001	0.000	0.001	0.001	0.001
50%	Debiased	0.003	0.001	0.003	0.002	0.005

Table 34: MSE Summary Statistics of β_{14} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.001	0.001	0.001	0.001	0.003
5%	Human Validation	0.014	0.003	0.015	0.010	0.017
5%	Debiased	3.543	23.671	0.099	0.055	2.287
10%	LLM	0.001	0.001	0.001	0.001	0.003
10%	Human Validation	0.007	0.001	0.007	0.005	0.009
10%	Debiased	0.037	0.013	0.036	0.020	0.057
25%	LLM	0.001	0.001	0.001	0.001	0.003
25%	Human Validation	0.003	0.001	0.003	0.002	0.003
25%	Debiased	0.016	0.005	0.015	0.009	0.026
50%	LLM	0.001	0.001	0.001	0.001	0.003
50%	Human Validation	0.001	0.000	0.001	0.001	0.002
50%	Debiased	0.012	0.004	0.012	0.007	0.018

Table 35: MSE Summary Statistics of β_{15} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.001	0.000	0.001	0.000	0.001
5%	Human Validation	0.008	0.002	0.009	0.005	0.011
5%	Debiased	0.018	0.006	0.018	0.009	0.029
10%	LLM	0.001	0.000	0.001	0.000	0.001
10%	Human Validation	0.004	0.001	0.004	0.002	0.005
10%	Debiased	0.008	0.003	0.009	0.004	0.013
25%	LLM	0.001	0.000	0.001	0.000	0.001
25%	Human Validation	0.002	0.000	0.002	0.001	0.002
25%	Debiased	0.004	0.001	0.004	0.002	0.006
50%	LLM	0.001	0.000	0.001	0.000	0.001
50%	Human Validation	0.001	0.000	0.001	0.000	0.001
50%	Debiased	0.003	0.001	0.004	0.002	0.005

Table 36: MSE Summary Statistics of β_{19} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.001	0.000	0.001	0.000	0.001
5%	Human Validation	0.006	0.001	0.006	0.004	0.008
5%	Debiased	0.061	0.369	0.015	0.008	0.038
10%	LLM	0.001	0.000	0.001	0.000	0.001
10%	Human Validation	0.003	0.001	0.003	0.002	0.004
10%	Debiased	0.008	0.003	0.007	0.004	0.012
25%	LLM	0.001	0.000	0.001	0.000	0.001
25%	Human Validation	0.001	0.000	0.001	0.001	0.001
25%	Debiased	0.004	0.001	0.003	0.002	0.006
50%	LLM	0.001	0.000	0.001	0.000	0.001
50%	Human Validation	0.001	0.000	0.001	0.000	0.001
50%	Debiased	0.003	0.001	0.003	0.002	0.005

Table 37: MSE Summary Statistics of β_{20} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.001	0.000	0.001	0.000	0.001
5%	Human Validation	0.007	0.002	0.008	0.005	0.009
5%	Debiased	0.009	0.006	0.009	0.004	0.015
10%	LLM	0.001	0.000	0.001	0.000	0.001
10%	Human Validation	0.004	0.001	0.004	0.002	0.005
10%	Debiased	0.004	0.001	0.004	0.002	0.007
25%	LLM	0.001	0.000	0.001	0.000	0.001
25%	Human Validation	0.001	0.000	0.002	0.001	0.002
25%	Debiased	0.002	0.001	0.002	0.001	0.004
50%	LLM	0.001	0.000	0.001	0.000	0.001
50%	Human Validation	0.001	0.000	0.001	0.000	0.001
50%	Debiased	0.002	0.001	0.002	0.001	0.003

Table 38: MSE Summary Statistics of β_{Other} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.000	0.000	0.000	0.000	0.000
5%	Human Validation	0.002	0.000	0.002	0.001	0.002
5%	Debiased	0.011	0.052	0.002	0.001	0.009
10%	LLM	0.000	0.000	0.000	0.000	0.000
10%	Human Validation	0.001	0.000	0.001	0.001	0.001
10%	Debiased	0.001	0.000	0.001	0.001	0.002
25%	LLM	0.000	0.000	0.000	0.000	0.000
25%	Human Validation	0.000	0.000	0.000	0.000	0.000
25%	Debiased	0.001	0.000	0.001	0.000	0.001
50%	LLM	0.000	0.000	0.000	0.000	0.000
50%	Human Validation	0.000	0.000	0.000	0.000	0.000
50%	Debiased	0.001	0.000	0.001	0.000	0.001

Table 39: MSE Summary Statistics of β_3 by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.001	0.001	0.001	0.001	0.003
GPT-3.5	5%	Human Validation	0.012	0.002	0.013	0.009	0.014
GPT-3.5	5%	Debiased	0.022	0.019	0.019	0.008	0.032
GPT-3.5	10%	LLM	0.001	0.001	0.002	0.001	0.003
GPT-3.5	10%	Human Validation	0.006	0.001	0.006	0.004	0.007
GPT-3.5	10%	Debiased	0.009	0.003	0.008	0.004	0.014
GPT-3.5	25%	LLM	0.001	0.001	0.001	0.001	0.003
GPT-3.5	25%	Human Validation	0.002	0.000	0.002	0.002	0.003
GPT-3.5	25%	Debiased	0.004	0.001	0.004	0.002	0.007
GPT-3.5	50%	LLM	0.001	0.001	0.002	0.001	0.003
GPT-3.5	50%	Human Validation	0.001	0.000	0.001	0.001	0.001
GPT-3.5	50%	Debiased	0.004	0.001	0.004	0.002	0.006
GPT-4o	5%	LLM	0.001	0.000	0.001	0.001	0.001
GPT-4o	5%	Human Validation	0.012	0.002	0.013	0.009	0.014
GPT-4o	5%	Debiased	0.009	0.002	0.009	0.006	0.011
GPT-4o	10%	LLM	0.001	0.000	0.001	0.001	0.001
GPT-4o	10%	Human Validation	0.006	0.001	0.006	0.004	0.007
GPT-4o	10%	Debiased	0.004	0.001	0.004	0.003	0.005
GPT-4o	25%	LLM	0.001	0.000	0.001	0.001	0.001
GPT-4o	25%	Human Validation	0.002	0.000	0.002	0.002	0.003
GPT-4o	25%	Debiased	0.002	0.000	0.003	0.002	0.003
GPT-4o	50%	LLM	0.001	0.000	0.001	0.001	0.001
GPT-4o	50%	Human Validation	0.001	0.000	0.001	0.001	0.001
GPT-4o	50%	Debiased	0.002	0.000	0.002	0.002	0.003

Table 40: MSE Summary Statistics of β_{14} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^{\top}\beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.002	0.001	0.002	0.001	0.003
GPT-3.5	5%	Human Validation	0.014	0.003	0.015	0.010	0.018
GPT-3.5	5%	Debiased	6.812	33.377	0.160	0.083	11.343
GPT-3.5	10%	LLM	0.002	0.001	0.002	0.001	0.003
GPT-3.5	10%	Human Validation	0.007	0.001	0.007	0.005	0.008
GPT-3.5	10%	Debiased	0.046	0.011	0.048	0.030	0.060
GPT-3.5	25%	LLM	0.002	0.001	0.002	0.001	0.003
GPT-3.5	25%	Human Validation	0.003	0.001	0.003	0.002	0.003
GPT-3.5	25%	Debiased	0.020	0.005	0.020	0.013	0.026
GPT-3.5	50%	LLM	0.002	0.001	0.002	0.001	0.003
GPT-3.5	50%	Human Validation	0.001	0.000	0.001	0.001	0.002
GPT-3.5	50%	Debiased	0.014	0.003	0.015	0.009	0.019
GPT-4o	5%	LLM	0.001	0.000	0.001	0.001	0.002
GPT-4o	5%	Human Validation	0.014	0.003	0.015	0.010	0.017
GPT-4o	5%	Debiased	0.273	0.804	0.080	0.050	0.778
GPT-4o	10%	LLM	0.001	0.000	0.001	0.001	0.002
GPT-4o	10%	Human Validation	0.007	0.001	0.007	0.005	0.009
GPT-4o	10%	Debiased	0.029	0.008	0.031	0.019	0.039
GPT-4o	25%	LLM	0.001	0.000	0.001	0.001	0.002
GPT-4o	25%	Human Validation	0.003	0.001	0.003	0.002	0.003
GPT-4o	25%	Debiased	0.013	0.003	0.013	0.009	0.017
GPT-4o	50%	LLM	0.001	0.000	0.001	0.001	0.002
GPT-4o	50%	Human Validation	0.001	0.000	0.001	0.001	0.002
GPT-4o	50%	Debiased	0.010	0.002	0.010	0.007	0.013

Table 41: MSE Summary Statistics of β_{15} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	5%	Human Validation	0.008	0.002	0.009	0.005	0.011
GPT-3.5	5%	Debiased	0.020	0.007	0.019	0.010	0.030
GPT-3.5	10%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	10%	Human Validation	0.004	0.001	0.004	0.002	0.005
GPT-3.5	10%	Debiased	0.009	0.003	0.009	0.004	0.013
GPT-3.5	25%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	25%	Human Validation	0.002	0.000	0.002	0.001	0.002
GPT-3.5	25%	Debiased	0.004	0.001	0.004	0.002	0.007
GPT-3.5	50%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	50%	Human Validation	0.001	0.000	0.001	0.000	0.001
GPT-3.5	50%	Debiased	0.004	0.001	0.004	0.002	0.005
GPT-4o	5%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-4o	5%	Human Validation	0.008	0.002	0.009	0.005	0.011
GPT-4o	5%	Debiased	0.016	0.005	0.018	0.009	0.023
GPT-4o	10%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-4o	10%	Human Validation	0.004	0.001	0.004	0.002	0.005
GPT-4o	10%	Debiased	0.008	0.002	0.008	0.004	0.011
GPT-4o	25%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-4o	25%	Human Validation	0.002	0.000	0.002	0.001	0.002
GPT-4o	25%	Debiased	0.004	0.001	0.004	0.002	0.005
GPT-4o	50%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-4o	50%	Human Validation	0.001	0.000	0.001	0.000	0.001
GPT-4o	50%	Debiased	0.003	0.001	0.004	0.002	0.005

Table 42: MSE Summary Statistics of β_{19} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	5%	Human Validation	0.006	0.001	0.006	0.004	0.008
GPT-3.5	5%	Debiased	0.108	0.521	0.021	0.012	0.042
GPT-3.5	10%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	10%	Human Validation	0.003	0.001	0.003	0.002	0.004
GPT-3.5	10%	Debiased	0.009	0.003	0.009	0.006	0.013
GPT-3.5	25%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	25%	Human Validation	0.001	0.000	0.001	0.001	0.001
GPT-3.5	25%	Debiased	0.004	0.001	0.005	0.003	0.006
GPT-3.5	50%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	50%	Human Validation	0.001	0.000	0.001	0.000	0.001
GPT-3.5	50%	Debiased	0.004	0.001	0.004	0.002	0.005
GPT-4o	5%	LLM	0.000	0.000	0.000	0.000	0.001
GPT-4o	5%	Human Validation	0.006	0.001	0.006	0.004	0.008
GPT-4o	5%	Debiased	0.013	0.005	0.013	0.008	0.018
GPT-4o	10%	LLM	0.000	0.000	0.000	0.000	0.001
GPT-4o	10%	Human Validation	0.003	0.001	0.003	0.002	0.004
GPT-4o	10%	Debiased	0.006	0.001	0.006	0.004	0.007
GPT-4o	25%	LLM	0.000	0.000	0.000	0.000	0.001
GPT-4o	25%	Human Validation	0.001	0.000	0.001	0.001	0.001
GPT-4o	25%	Debiased	0.003	0.001	0.003	0.002	0.004
GPT-4o	50%	LLM	0.000	0.000	0.000	0.000	0.001
GPT-4o	50%	Human Validation	0.001	0.000	0.001	0.000	0.001
GPT-4o	50%	Debiased	0.002	0.001	0.003	0.002	0.003

Table 43: MSE Summary Statistics of β_{20} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	5%	Human Validation	0.007	0.002	0.008	0.005	0.009
GPT-3.5	5%	Debiased	0.011	0.008	0.010	0.005	0.016
GPT-3.5	10%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	10%	Human Validation	0.004	0.001	0.004	0.002	0.005
GPT-3.5	10%	Debiased	0.005	0.002	0.005	0.003	0.008
GPT-3.5	25%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	25%	Human Validation	0.001	0.000	0.002	0.001	0.002
GPT-3.5	25%	Debiased	0.003	0.001	0.003	0.001	0.004
GPT-3.5	50%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	50%	Human Validation	0.001	0.000	0.001	0.000	0.001
GPT-3.5	50%	Debiased	0.002	0.001	0.002	0.001	0.003
GPT-4o	5%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-4o	5%	Human Validation	0.007	0.002	0.008	0.005	0.009
GPT-4o	5%	Debiased	0.007	0.002	0.008	0.004	0.010
GPT-4o	10%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-4o	10%	Human Validation	0.004	0.001	0.004	0.002	0.005
GPT-4o	10%	Debiased	0.004	0.001	0.004	0.002	0.005
GPT-4o	25%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-4o	25%	Human Validation	0.001	0.000	0.002	0.001	0.002
GPT-4o	25%	Debiased	0.002	0.001	0.002	0.001	0.003
GPT-4o	50%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-4o	50%	Human Validation	0.001	0.000	0.001	0.000	0.001
GPT-4o	50%	Debiased	0.002	0.001	0.002	0.001	0.002

Table 44: MSE Summary Statistics of β_{Other} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.000	0.000	0.000	0.000	0.000
GPT-3.5	5%	Human Validation	0.002	0.000	0.002	0.001	0.002
GPT-3.5	5%	Debiased	0.019	0.073	0.003	0.002	0.047
GPT-3.5	10%	LLM	0.000	0.000	0.000	0.000	0.000
GPT-3.5	10%	Human Validation	0.001	0.000	0.001	0.001	0.001
GPT-3.5	10%	Debiased	0.001	0.000	0.001	0.001	0.002
GPT-3.5	25%	LLM	0.000	0.000	0.000	0.000	0.000
GPT-3.5	25%	Human Validation	0.000	0.000	0.000	0.000	0.000
GPT-3.5	25%	Debiased	0.001	0.000	0.001	0.000	0.001
GPT-3.5	50%	LLM	0.000	0.000	0.000	0.000	0.000
GPT-3.5	50%	Human Validation	0.000	0.000	0.000	0.000	0.000
GPT-3.5	50%	Debiased	0.001	0.000	0.001	0.000	0.001
GPT-4o	5%	LLM	0.000	0.000	0.000	0.000	0.000
GPT-4o	5%	Human Validation	0.002	0.000	0.002	0.001	0.002
GPT-4o	5%	Debiased	0.002	0.002	0.002	0.001	0.005
GPT-4o	10%	LLM	0.000	0.000	0.000	0.000	0.000
GPT-4o	10%	Human Validation	0.001	0.000	0.001	0.001	0.001
GPT-4o	10%	Debiased	0.001	0.000	0.001	0.001	0.001
GPT-4o	25%	LLM	0.000	0.000	0.000	0.000	0.000
GPT-4o	25%	Human Validation	0.000	0.000	0.000	0.000	0.000
GPT-4o	25%	Debiased	0.000	0.000	0.001	0.000	0.001
GPT-4o	50%	LLM	0.000	0.000	0.000	0.000	0.000
GPT-4o	50%	Human Validation	0.000	0.000	0.000	0.000	0.000
GPT-4o	50%	Debiased	0.000	0.000	0.000	0.000	0.001

Table 45: Coverage Summary Statistics of β_3 by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.867	0.096	0.911	0.689	0.952
5%	Human Validation	0.919	0.008	0.919	0.906	0.931
5%	Debiased	0.920	0.011	0.921	0.904	0.939
10%	LLM	0.866	0.097	0.913	0.679	0.947
10%	Human Validation	0.935	0.008	0.935	0.923	0.947
10%	Debiased	0.940	0.007	0.940	0.928	0.952
25%	LLM	0.865	0.097	0.909	0.692	0.948
25%	Human Validation	0.944	0.008	0.944	0.931	0.954
25%	Debiased	0.945	0.007	0.946	0.934	0.955
50%	LLM	0.865	0.099	0.913	0.653	0.947
50%	Human Validation	0.949	0.007	0.948	0.939	0.959
50%	Debiased	0.948	0.007	0.949	0.937	0.957

Table 46: Coverage Summary Statistics of β_{14} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^{\top}\beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.685	0.225	0.764	0.252	0.932
5%	Human Validation	0.917	0.009	0.917	0.901	0.930
5%	Debiased	0.938	0.010	0.939	0.920	0.953
10%	LLM	0.689	0.226	0.770	0.242	0.937
10%	Human Validation	0.933	0.008	0.934	0.918	0.945
10%	Debiased	0.940	0.008	0.942	0.926	0.951
25%	LLM	0.686	0.225	0.759	0.237	0.938
25%	Human Validation	0.943	0.008	0.944	0.929	0.955
25%	Debiased	0.946	0.007	0.946	0.937	0.957
50%	LLM	0.690	0.225	0.766	0.255	0.935
50%	Human Validation	0.947	0.007	0.948	0.936	0.959
50%	Debiased	0.948	0.007	0.949	0.937	0.958

Table 47: Coverage Summary Statistics of β_{15} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^{\top}\beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.925	0.032	0.936	0.879	0.957
5%	Human Validation	0.932	0.009	0.931	0.919	0.946
5%	Debiased	0.932	0.010	0.933	0.919	0.950
10%	LLM	0.926	0.030	0.938	0.876	0.953
10%	Human Validation	0.940	0.008	0.941	0.927	0.951
10%	Debiased	0.941	0.008	0.941	0.928	0.955
25%	LLM	0.925	0.033	0.938	0.868	0.953
25%	Human Validation	0.946	0.007	0.946	0.934	0.958
25%	Debiased	0.947	0.007	0.947	0.934	0.957
50%	LLM	0.926	0.031	0.933	0.883	0.956
50%	Human Validation	0.947	0.006	0.947	0.938	0.958
50%	Debiased	0.947	0.007	0.947	0.934	0.957

Table 48: Coverage Summary Statistics of β_{19} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^{\top}\beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.891	0.065	0.915	0.740	0.954
5%	Human Validation	0.937	0.009	0.938	0.926	0.953
5%	Debiased	0.944	0.009	0.944	0.931	0.956
10%	LLM	0.890	0.066	0.912	0.750	0.949
10%	Human Validation	0.943	0.008	0.943	0.930	0.954
10%	Debiased	0.947	0.008	0.947	0.934	0.957
25%	LLM	0.894	0.064	0.915	0.737	0.954
25%	Human Validation	0.947	0.007	0.947	0.938	0.958
25%	Debiased	0.947	0.008	0.948	0.935	0.959
50%	LLM	0.889	0.066	0.913	0.735	0.947
50%	Human Validation	0.951	0.007	0.950	0.940	0.961
50%	Debiased	0.947	0.007	0.948	0.935	0.957

Table 49: Coverage Summary Statistics of β_{20} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^{\top}\beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.901	0.062	0.923	0.784	0.956
5%	Human Validation	0.938	0.007	0.938	0.926	0.950
5%	Debiased	0.938	0.008	0.938	0.926	0.951
10%	LLM	0.902	0.056	0.920	0.785	0.953
10%	Human Validation	0.943	0.007	0.942	0.932	0.956
10%	Debiased	0.945	0.007	0.944	0.934	0.956
25%	LLM	0.903	0.058	0.925	0.783	0.952
25%	Human Validation	0.948	0.007	0.948	0.937	0.959
25%	Debiased	0.947	0.007	0.947	0.937	0.959
50%	LLM	0.904	0.055	0.923	0.796	0.954
50%	Human Validation	0.950	0.007	0.951	0.938	0.960
50%	Debiased	0.949	0.007	0.948	0.939	0.961

Table 50: Coverage Summary Statistics of β_{Other} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^{\top}\beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.836	0.133	0.886	0.570	0.950
5%	Human Validation	0.949	0.007	0.949	0.937	0.961
5%	Debiased	0.954	0.007	0.954	0.943	0.966
10%	LLM	0.841	0.133	0.891	0.550	0.955
10%	Human Validation	0.950	0.007	0.950	0.939	0.960
10%	Debiased	0.951	0.008	0.952	0.938	0.962
25%	LLM	0.839	0.133	0.887	0.581	0.949
25%	Human Validation	0.949	0.007	0.950	0.938	0.958
25%	Debiased	0.948	0.007	0.950	0.935	0.959
50%	LLM	0.840	0.129	0.891	0.561	0.951
50%	Human Validation	0.951	0.007	0.950	0.940	0.962
50%	Debiased	0.949	0.007	0.949	0.939	0.960

Table 51: Coverage Summary Statistics of β_3 by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.812	0.108	0.804	0.642	0.938
GPT-3.5	5%	Human Validation	0.918	0.008	0.917	0.905	0.931
GPT-3.5	5%	Debiased	0.926	0.010	0.927	0.913	0.941
GPT-3.5	10%	LLM	0.811	0.110	0.810	0.636	0.934
GPT-3.5	10%	Human Validation	0.935	0.007	0.937	0.923	0.945
GPT-3.5	10%	Debiased	0.940	0.009	0.940	0.925	0.955
GPT-3.5	25%	LLM	0.810	0.109	0.800	0.623	0.936
GPT-3.5	25%	Human Validation	0.942	0.007	0.942	0.929	0.955
GPT-3.5	25%	Debiased	0.946	0.006	0.946	0.934	0.955
GPT-3.5	50%	LLM	0.809	0.113	0.820	0.621	0.937
GPT-3.5	50%	Human Validation	0.949	0.008	0.948	0.938	0.961
GPT-3.5	50%	Debiased	0.946	0.007	0.946	0.937	0.956
GPT-4o	5%	LLM	0.923	0.030	0.932	0.874	0.958
GPT-4o	5%	Human Validation	0.920	0.008	0.920	0.908	0.933
GPT-4o	5%	Debiased	0.914	0.009	0.913	0.901	0.928
GPT-4o	10%	LLM	0.922	0.027	0.933	0.871	0.950
GPT-4o	10%	Human Validation	0.935	0.008	0.935	0.923	0.948
GPT-4o	10%	Debiased	0.940	0.005	0.940	0.929	0.947
GPT-4o	25%	LLM	0.921	0.030	0.934	0.875	0.951
GPT-4o	25%	Human Validation	0.946	0.007	0.946	0.933	0.954
GPT-4o	25%	Debiased	0.945	0.007	0.946	0.933	0.954
GPT-4o	50%	LLM	0.921	0.027	0.929	0.874	0.948
GPT-4o	50%	Human Validation	0.948	0.007	0.948	0.939	0.958
GPT-4o	50%	Debiased	0.949	0.006	0.951	0.940	0.957

Table 52: Coverage Summary Statistics of β_{14} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.596	0.255	0.650	0.178	0.903
GPT-3.5	5%	Human Validation	0.917	0.010	0.917	0.901	0.933
GPT-3.5	5%	Debiased	0.944	0.006	0.944	0.936	0.956
GPT-3.5	10%	LLM	0.599	0.255	0.667	0.200	0.913
GPT-3.5	10%	Human Validation	0.932	0.008	0.932	0.916	0.944
GPT-3.5	10%	Debiased	0.942	0.010	0.943	0.926	0.953
GPT-3.5	25%	LLM	0.599	0.256	0.659	0.172	0.915
GPT-3.5	25%	Human Validation	0.943	0.008	0.944	0.931	0.953
GPT-3.5	25%	Debiased	0.946	0.009	0.947	0.933	0.959
GPT-3.5	50%	LLM	0.602	0.255	0.649	0.184	0.921
GPT-3.5	50%	Human Validation	0.947	0.007	0.948	0.937	0.959
GPT-3.5	50%	Debiased	0.948	0.006	0.949	0.939	0.957
GPT-4o	5%	LLM	0.775	0.147	0.777	0.540	0.941
GPT-4o	5%	Human Validation	0.917	0.008	0.917	0.905	0.928
GPT-4o	5%	Debiased	0.933	0.010	0.935	0.920	0.946
GPT-4o	10%	LLM	0.779	0.150	0.792	0.524	0.950
GPT-4o	10%	Human Validation	0.934	0.008	0.935	0.919	0.945
GPT-4o	10%	Debiased	0.939	0.007	0.940	0.928	0.949
GPT-4o	25%	LLM	0.774	0.146	0.786	0.536	0.940
GPT-4o	25%	Human Validation	0.943	0.008	0.944	0.928	0.955
GPT-4o	25%	Debiased	0.946	0.006	0.944	0.938	0.957
GPT-4o	50%	LLM	0.777	0.148	0.793	0.537	0.939
GPT-4o	50%	Human Validation	0.947	0.008	0.948	0.934	0.958
GPT-4o	50%	Debiased	0.948	0.008	0.949	0.936	0.959

Table 53: Coverage Summary Statistics of β_{15} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.914	0.038	0.923	0.826	0.954
GPT-3.5	5%	Human Validation	0.932	0.008	0.932	0.923	0.944
GPT-3.5	5%	Debiased	0.938	0.008	0.938	0.927	0.951
GPT-3.5	10%	LLM	0.915	0.036	0.926	0.853	0.950
GPT-3.5	10%	Human Validation	0.940	0.008	0.942	0.926	0.953
GPT-3.5	10%	Debiased	0.943	0.008	0.944	0.930	0.956
GPT-3.5	25%	LLM	0.913	0.039	0.923	0.832	0.950
GPT-3.5	25%	Human Validation	0.946	0.007	0.946	0.934	0.959
GPT-3.5	25%	Debiased	0.947	0.005	0.947	0.939	0.955
GPT-3.5	50%	LLM	0.915	0.037	0.921	0.836	0.954
GPT-3.5	50%	Human Validation	0.947	0.005	0.945	0.939	0.954
GPT-3.5	50%	Debiased	0.948	0.006	0.947	0.936	0.958
GPT-4o	5%	LLM	0.936	0.019	0.940	0.892	0.957
GPT-4o	5%	Human Validation	0.932	0.010	0.931	0.916	0.947
GPT-4o	5%	Debiased	0.927	0.009	0.926	0.912	0.941
GPT-4o	10%	LLM	0.937	0.017	0.943	0.903	0.953
GPT-4o	10%	Human Validation	0.940	0.007	0.940	0.928	0.948
GPT-4o	10%	Debiased	0.939	0.009	0.940	0.925	0.952
GPT-4o	25%	LLM	0.937	0.019	0.942	0.901	0.958
GPT-4o	25%	Human Validation	0.947	0.008	0.947	0.934	0.956
GPT-4o	25%	Debiased	0.946	0.008	0.946	0.933	0.958
GPT-4o	50%	LLM	0.937	0.019	0.943	0.895	0.956
GPT-4o	50%	Human Validation	0.947	0.006	0.948	0.938	0.959
GPT-4o	50%	Debiased	0.945	0.007	0.948	0.933	0.955

Table 54: Coverage Summary Statistics of β_{19} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.865	0.078	0.899	0.715	0.948
GPT-3.5	5%	Human Validation	0.937	0.007	0.938	0.928	0.947
GPT-3.5	5%	Debiased	0.945	0.008	0.946	0.932	0.956
GPT-3.5	10%	LLM	0.866	0.078	0.897	0.719	0.944
GPT-3.5	10%	Human Validation	0.943	0.009	0.943	0.930	0.957
GPT-3.5	10%	Debiased	0.945	0.009	0.946	0.934	0.957
GPT-3.5	25%	LLM	0.868	0.078	0.893	0.715	0.949
GPT-3.5	25%	Human Validation	0.948	0.007	0.948	0.938	0.958
GPT-3.5	25%	Debiased	0.948	0.008	0.948	0.936	0.960
GPT-3.5	50%	LLM	0.864	0.081	0.895	0.717	0.943
GPT-3.5	50%	Human Validation	0.951	0.007	0.950	0.940	0.963
GPT-3.5	50%	Debiased	0.946	0.008	0.946	0.934	0.958
GPT-4o	5%	LLM	0.917	0.034	0.927	0.853	0.957
GPT-4o	5%	Human Validation	0.938	0.010	0.939	0.917	0.953
GPT-4o	5%	Debiased	0.942	0.009	0.941	0.928	0.956
GPT-4o	10%	LLM	0.914	0.038	0.928	0.848	0.950
GPT-4o	10%	Human Validation	0.943	0.007	0.942	0.932	0.952
GPT-4o	10%	Debiased	0.948	0.008	0.948	0.935	0.958
GPT-4o	25%	LLM	0.919	0.030	0.929	0.862	0.956
GPT-4o	25%	Human Validation	0.946	0.007	0.945	0.936	0.956
GPT-4o	25%	Debiased	0.947	0.008	0.947	0.935	0.958
GPT-4o	50%	LLM	0.914	0.031	0.921	0.860	0.950
GPT-4o	50%	Human Validation	0.951	0.006	0.950	0.941	0.960
GPT-4o	50%	Debiased	0.948	0.005	0.948	0.938	0.954

Table 55: Coverage Summary Statistics of β_{20} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.913	0.053	0.930	0.822	0.956
GPT-3.5	5%	Human Validation	0.938	0.006	0.938	0.930	0.946
GPT-3.5	5%	Debiased	0.941	0.007	0.942	0.929	0.953
GPT-3.5	10%	LLM	0.911	0.049	0.927	0.813	0.953
GPT-3.5	10%	Human Validation	0.942	0.008	0.941	0.932	0.956
GPT-3.5	10%	Debiased	0.944	0.008	0.941	0.934	0.958
GPT-3.5	25%	LLM	0.912	0.051	0.930	0.821	0.952
GPT-3.5	25%	Human Validation	0.946	0.007	0.946	0.937	0.956
GPT-3.5	25%	Debiased	0.948	0.006	0.948	0.938	0.960
GPT-3.5	50%	LLM	0.912	0.050	0.929	0.818	0.955
GPT-3.5	50%	Human Validation	0.950	0.006	0.951	0.941	0.959
GPT-3.5	50%	Debiased	0.948	0.008	0.946	0.937	0.963
GPT-4o	5%	LLM	0.889	0.068	0.902	0.796	0.952
GPT-4o	5%	Human Validation	0.937	0.009	0.937	0.924	0.950
GPT-4o	5%	Debiased	0.936	0.008	0.936	0.924	0.947
GPT-4o	10%	LLM	0.893	0.062	0.915	0.794	0.953
GPT-4o	10%	Human Validation	0.944	0.006	0.943	0.935	0.955
GPT-4o	10%	Debiased	0.945	0.006	0.946	0.934	0.955
GPT-4o	25%	LLM	0.893	0.063	0.914	0.792	0.951
GPT-4o	25%	Human Validation	0.949	0.007	0.950	0.937	0.962
GPT-4o	25%	Debiased	0.946	0.007	0.945	0.936	0.956
GPT-4o	50%	LLM	0.895	0.060	0.917	0.796	0.951
GPT-4o	50%	Human Validation	0.950	0.008	0.950	0.936	0.963
GPT-4o	50%	Debiased	0.949	0.007	0.949	0.940	0.959

Table 56: Coverage Summary Statistics of β_{Other} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.769	0.156	0.797	0.503	0.944
GPT-3.5	5%	Human Validation	0.950	0.007	0.950	0.938	0.963
GPT-3.5	5%	Debiased	0.956	0.007	0.957	0.947	0.969
GPT-3.5	10%	LLM	0.773	0.157	0.805	0.508	0.942
GPT-3.5	10%	Human Validation	0.951	0.007	0.951	0.940	0.961
GPT-3.5	10%	Debiased	0.952	0.007	0.952	0.941	0.963
GPT-3.5	25%	LLM	0.774	0.156	0.804	0.502	0.947
GPT-3.5	25%	Human Validation	0.950	0.007	0.951	0.938	0.959
GPT-3.5	25%	Debiased	0.948	0.007	0.948	0.935	0.960
GPT-3.5	50%	LLM	0.774	0.151	0.796	0.522	0.942
GPT-3.5	50%	Human Validation	0.950	0.006	0.951	0.940	0.960
GPT-3.5	50%	Debiased	0.948	0.008	0.949	0.936	0.961
GPT-4o	5%	LLM	0.903	0.047	0.919	0.813	0.956
GPT-4o	5%	Human Validation	0.947	0.007	0.947	0.938	0.959
GPT-4o	5%	Debiased	0.952	0.007	0.952	0.943	0.965
GPT-4o	10%	LLM	0.908	0.046	0.928	0.829	0.956
GPT-4o	10%	Human Validation	0.949	0.007	0.950	0.938	0.958
GPT-4o	10%	Debiased	0.950	0.008	0.952	0.936	0.962
GPT-4o	25%	LLM	0.905	0.049	0.923	0.819	0.951
GPT-4o	25%	Human Validation	0.948	0.006	0.949	0.939	0.957
GPT-4o	25%	Debiased	0.948	0.008	0.951	0.935	0.958
GPT-4o	50%	LLM	0.907	0.045	0.920	0.832	0.958
GPT-4o	50%	Human Validation	0.951	0.007	0.949	0.941	0.964
GPT-4o	50%	Debiased	0.950	0.006	0.949	0.941	0.959